6

Purchasing Power Parity and Floating Exchange Rate Experience

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6.1 Introduction

In Chapter 1, we looked at what exactly the foreign exchange market is, introduced a number of differing exchange rate concepts and examined a simple current account model of exchange rate determination. In this chapter we look at one of the earliest and simplest models of exchange rate determination known as purchasing power parity (PPP) theory. An understanding of PPP is essential to the study of international finance. PPP theory has been advocated as a satisfactory model of exchange rate determination in its own right, and also provides a point of reference for the long-run exchange rate in many of the modern exchange rate theories which we examine in later chapters.

Having looked at PPP theory, we proceed to examine how well-suited this theory is to explaining actual exchange rate behaviour since the adoption of generalized floating

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In 1973. As we shall see, PPP theory does not provide an adequate explanation of some of the observed features of floating exchange rates. Some of the possible explanations for the failure of PPP are then discussed.

6.2 Purchasing power parity theory and the law of one price

PPP is generally attributed to Gustav Cassell's writings in the 1920s, although its intellectual origins date back to the writings of the nineteenth-century British economist David Ricardo. The basic concept underlying PPP theory is that arbitrage forces will lead to the equalization of goods prices internationally once the prices of goods are measured in the same currency. As such the theory represents an application of the 'law of one price'.

The law of one price

The law of one price simply says that in the presence of a competitive market structure and the absence of transport costs and other barriers to trade, identical products which are sold in different markets will sell at the same price when expressed in terms of a common currency.

The law of one price is based on the idea of perfect goods arbitrage. Arbitrage occurs where economic agents exploit price differences so as to provide a riskless profit. For example, if a car costs £15,000 in the UK and the identical model costs $30,000 in the USA, then according to the law of one price the exchange rate should be £15,000/$30,000, which is £0.50/$1. If the exchange rate were higher than this, say at £0.60/$1, then it would pay a US resident to purchase a car in the UK because with $25,000 he would obtain £15,000 which could then be used to purchase a car in the UK saving $5,000 compared to purchasing in the USA. According to the law of one price, US residents will exploit this arbitrage possibility and start purchasing pounds and selling dollars. Such a process will continue until the pound appreciates to £0.50/$1 at which point arbitrage profit opportunities are eliminated. Conversely, if the exchange rate is £0.4/$1 then a UK car would cost a US resident £15,000/0.4 = $37,500, while a US car would cost UK residents $30,000 × 0.4 = £12,000, the pound is overvalued. Hence, US residents will not buy UK cars and UK residents will buy US cars (saving £3,000) so the pound will depreciate on the foreign exchange market to its PPP value of £0.50/$1.

The proponents of PPP argue that the exchange rate must adjust to ensure that the 'law of one price' which applies only to individual goods, also holds internationally for identical bundles of goods.

6.3 Absolute and relative PPP

Purchasing power parity theory comes in two forms; one is based on a strict interpretation of the law of one price and is termed absolute purchasing power parity, while the other is a 'weaker' variation known as relative purchasing power parity. We shall first examine the absolute version of the theory and then consider the relative version.

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Absolute PPP

The absolute version of PPP holds that if one takes a bundle of goods in one country and compares the price of that bundle with an identical bundle of goods sold in a foreign country converted by the exchange rate into a common currency of measurement, then the prices will be equal. For example, if a bundle of goods costs £100 in the UK and the same bundle costs $200 in the USA, then the exchange rate defined as pounds per dollar will be £100/$200 = £0.50/$1. Algebraically, the absolute version of PPP can be stated as:

$$S = \frac{P}{P^*} \tag{6.1}$$

where S is the exchange rate defined as domestic currency units per unit of foreign currency, P is the price of a bundle of goods expressed in the domestic currency, and P^* is the price of an identical bundle of goods in the foreign country expressed in terms of the foreign currency.

According to absolute PPP, a rise in the home price level relative to the foreign price level will lead to a proportional depreciation of the home currency against the foreign currency. In our example, if the price of the UK bundle rises to £160 while the price of the US bundle remains at $200, then according to absolute PPP the pound will depreciate to £0.80/$1.

Relative PPP

The absolute version of PPP is, even proponents of the theory generally acknowledge, unlikely to hold precisely because of the existence of transport costs, imperfect information and the distorting effects of tariff and non-tariff barriers to trade.

Nonetheless, it is argued that a weaker form of PPP known as relative purchasing power parity can be expected to hold even in the presence of such distortions. Put simply, the relative version of PPP theory argues that the exchange rate will adjust by the amount of the inflation differential between two economies. Algebraically this is expressed as:

$$\%\Delta S = \%\Delta P - \%\Delta P^* \tag{6.2}$$

where $\%\Delta S$ is the percentage change in the exchange rate, $\%\Delta P$ is the domestic inflation rate, and $\%\Delta P^*$ is the foreign inflation rate.

According to the relative version of PPP, if the inflation rate in the UK is 10% whilst that in the United States is 4%, the pounds per dollar exchange rate should be expected to depreciate by approximately 6%. The absolute version of PPP does not have to hold for this to be the case. For example, the exchange rate may be £0.5/$1 while the UK bundle of goods costs £120 and the US bundle of identical goods costs $200 so that absolute PPP is not holding (this requires a rate of £0.60/$1). But if UK prices go up 10% to £132 and the US bundle goes up 4% to $212, the relative version of PPP predicts the pound will depreciate 6% to £0.53/$1 (even though absolute PPP requires £0.622/$1=£132/$212).

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6.3 A generalized version of PPP

One of the major problems with PPP theory as we have so far examined it, is that it suggests that PPP holds for all types of goods. However, a more generalized version of PPP that provides some useful insights makes a distinction between traded and non-traded goods. Traded goods are goods that are susceptible to the rigours of international competition, be they exports or import-competing industries such as most manufactured goods. Non-traded goods are those that cannot be traded internationally at a profit, such as houses and certain services such as a haircut, or restaurant food.

The point of the traded/non-traded goods distinction is that on *a priori* grounds PPP is more likely to hold for traded than for non-traded goods. This is because the price of traded goods will tend to be kept in line by international competition, while the price of non-traded goods will be determined predominately by domestic supply and demand considerations. For example, if a car costs £15,000 in the UK and $30,000 in the USA, arbitrage will tend to keep the pound–dollar rate at £0.50/$1. However, if the price of a house costs £150,000 in the UK and $80,000 in the USA and the exchange rate is £0.50/$1, arbitrage forces do not easily come into play (unless fed-up UK citizens emigrate to America pushing up US house prices and lowering UK prices). Similarly, if a haircut costs £10 in the UK but $10 in the USA and the exchange rate is £0.50/$1, only insane people in the UK will travel to the USA for a haircut knowing that they can save £5 because of the time and transport costs involved.

We now consider the importance of the tradables/non-tradables distinction for PPP when aggregate price indices made up of both tradables and non-tradables are considered. In the first instance, we assume that PPP holds for tradables goods, which means:

$$P_T = S P_{T^*} \tag{6.3}$$

where S is the exchange rate defined as domestic currency units per unit of foreign currency, P_T is the price of traded goods in the domestic country measured in terms of the domestic currency, and P_{T^*} is the price of traded goods in the foreign country measured in terms of the foreign currency.

The aggregate price index (P_I) for the domestic economy is made up of a weighted average of the price of both tradable (P_T) and non-tradable goods (P_N) priced in the domestic currency. Likewise, the foreign aggregate price index (P_{I^*}) is made up of a weighted average of the price on tradables (P_{T^*}) and non-tradables (P_{N^*}) priced in the foreign currency. This gives:

$$P_I = \alpha P_N + (1 - \alpha) P_T \tag{6.4}$$

where α is the proportion of non-traded goods in the domestic price index;

$$P_{I^*} = \beta P_{N^*} + (1 - \beta) P_{T^*} \tag{6.5}$$

where β is the proportion of non-traded goods in the foreign price index. Dividing equation (6.4) by (6.5) we obtain:

$$\frac{P_I}{P_{I^*}} = \frac{\alpha P_N + (1 - \alpha) P_T}{\beta P_{N^*} + (1 - \beta) P_{T^*}} \tag{6.6}$$

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If we divide the numerator by P_T and the denominator by SP_{T^*}, which, because of the assumption of PPP for tradable goods, are equivalent expressions (see equation 6.3) we obtain:

$$\frac{P_I}{P_I^*} = S \times \left[\frac{\alpha(P_N/P_T) + (1-\alpha)}{\beta(P_{N^*}/P_{T^*}) + (1-\beta)} \right] \tag{6.7}$$

This can be rearranged to give the solution for the exchange rate as:

$$S = \frac{P_I}{P_{I^*}} \times \left[\frac{\beta(P_{N^*}/P_{T^*}) + (1-\beta)}{\alpha(P_N/P_T) + (1-\alpha)} \right] \tag{6.8}$$

Equation (6.8) is an important modification to our simple PPP equation, because PPP no longer necessarily holds in terms of aggregate price indices due to the multiplicative term on the right-hand side. Furthermore, the equation suggests that the relative price of non-tradables relative to tradables will influence the exchange rate. If the domestic price of non-tradables rises relative to tradables this will lead to an appreciation (fall) of the home currency. The reason is that PPP holds only in terms of tradable goods. A rise in the relative price of non-tradables while keeping the aggregate price index constant implies that the price of tradables must have fallen which will then induce an appreciation of the exchange rate (to maintain PPP for tradable goods) even though the aggregate price index has remained unchanged. Conversely, a rise in the price of tradables while holding the aggregate price index constant leads to a depreciation of the exchange rate to maintain PPP for tradable goods.

The tradable/non-tradable distinction serves as a warning when testing for PPP. Testing for PPP using price indices based on tradable goods prices is likely to lead to better results than when using aggregate price indices made up of both types of goods. Exchange rate movements induced by changes in the relative prices between tradable and non-tradable goods represent real exchange rate changes. Among the factors that can lead to such relative price changes are differing rates of productivity in the traded and non-traded sectors of the economy, and changing consumer demand patterns.

6.5 Measurement problems in testing for PPP

Many of the proponents of PPP argued prior to the adoption of floating exchange rates that exchange rate changes would be in line with levels predicted by purchasing power parity theory. Before examining some of the empirical evidence on PPP theory, it is worth considering some of the practical problems involved in testing for PPP. One of the major problems is to decide whether or not the theory is supposed to be applicable to both traded goods and non-traded goods, or applicable to only one of those categories. At first sight PPP theory seems more readily applicable to traded goods, but some authors have argued that the distinction between the two categories is fuzzy and there are mechanisms linking both traded and non-traded goods prices. For example, some traded goods are used as inputs into the production of non-traded goods, and *vice-versa* (for example shop rents differ in price from the USA compared to say Mexico).

The argument over whether or not PPP should be applied to traded goods only or

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whether a more general price index made up of both traded and non-traded goods should be used is important for the empirical testing of PPP theory. If the theory is supposed to be applicable to traded goods only, then the price index used for testing the theory needs to be made up only of traded goods. Conversely, if the theory is applicable to both traded and non-traded goods then a more general price index should be employed. In practice, researchers who test PPP theory for traded goods typically use wholesale or manufacturing price indices which are normally dominated by traded goods, while if the test involves both traded and non-traded goods then consumer price indices which weight both classes of goods are generally used. An overall problem facing researchers, whichever price index they decide to employ, is that PPP is only expected to hold for similar baskets of goods, but national price indices typically attach different weights to different classes of goods. For instance, consumer price indices in underdeveloped economies typically have a high weighting for food, while those in developed countries have a lower weighting for food and a higher weighting for consumer goods.

Another statistical problem in testing for PPP is that the base period for the test should ideally be one where PPP held approximately. In addition, there are divergences of view over the time span during which PPP can be expected to assert itself; a strong version of PPP would suggest it holds on a monthly basis whereas progressively weaker versions would argue that it can be expected to hold only on a quarterly, six monthly or yearly and beyond basis. Bearing in mind some of these practical problems we proceed to look at some of the empirical evidence on PPP theory.

Empirical evidence on PPP

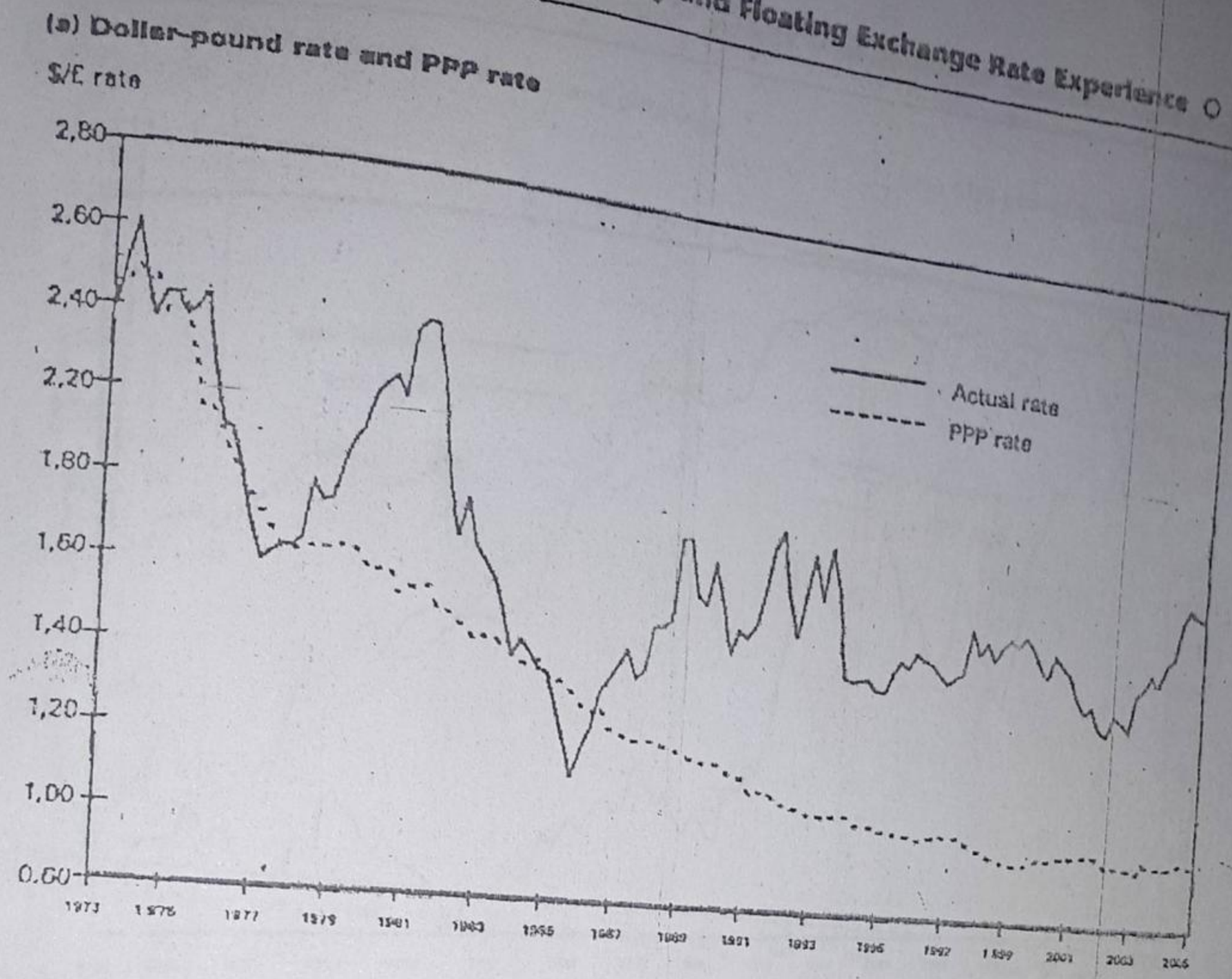
There are a variety of methods of testing for PPP, including graphical evidence, simplistic data analysis and more sophisticated econometric evidence. We first look at the graphical evidence.

Graphical evidence on PPP

Figure 6.1 plots the actual exchange rate and the exchange rate that would have maintained PPP, for various currencies, and shows that the exchange rate has diverged considerably from that suggested by purchasing power parity.

In Figure 6.1(a) it can be seen that PPP does not do at all well in tracking the dollar–pound rate – it performs reasonably well up to 1977 but between mid-1976 and mid-1981 there is a dramatic appreciation of the pound while PPP would have predicted a depreciation (due to higher UK inflation). A massive dollar appreciation after 1981 leads to the restoration of PPP in early-1984. Thereafter the pound has a brief period of undervaluation in relation to PPP and from late 1985 to 2005 the pound becomes rather overvalued in relation to PPP. Neither is the deutschmark–dollar rate (Figure 6.1(b)) explained by PPP, with the dollar generally undervalued in relation to PPP up to early 1981. Thereafter the dollar becomes substantially overvalued in relation to PPP up until mid-1986 when it once again becomes undervalued in relation to PPP. Again with the yen–dollar (Figure 6.1(c)) there are sustained and marked departures from PPP. Figure 6.1(d) shows that PPP does a poor job at tracking the yen–deutschmark parity.

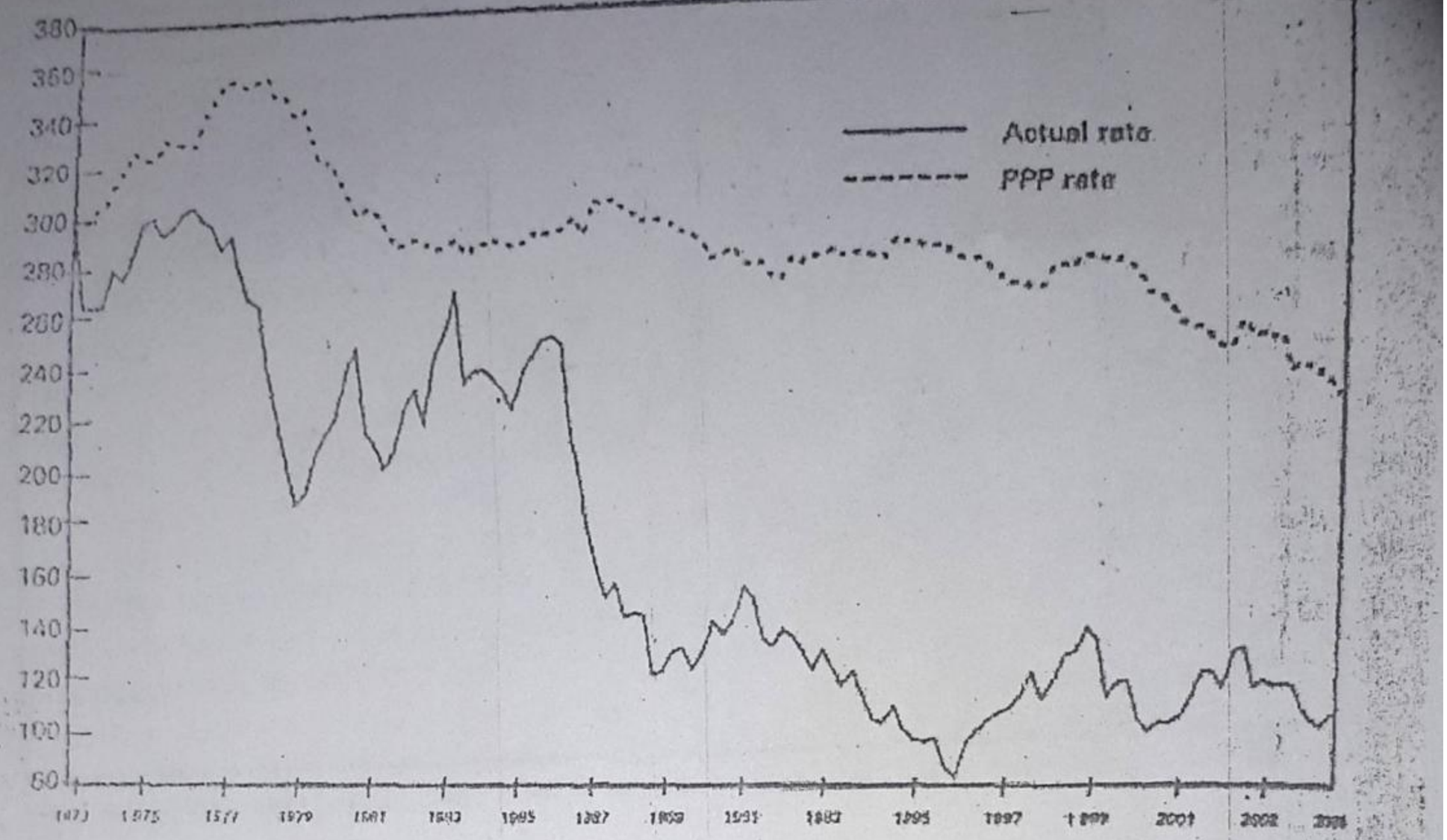
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The actual exchange rate and the PPP exchange rate for different currency pairs

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(c) Yen-dollar rate and PPP rate



(d) Yen-deutschmark rate and PPP rate

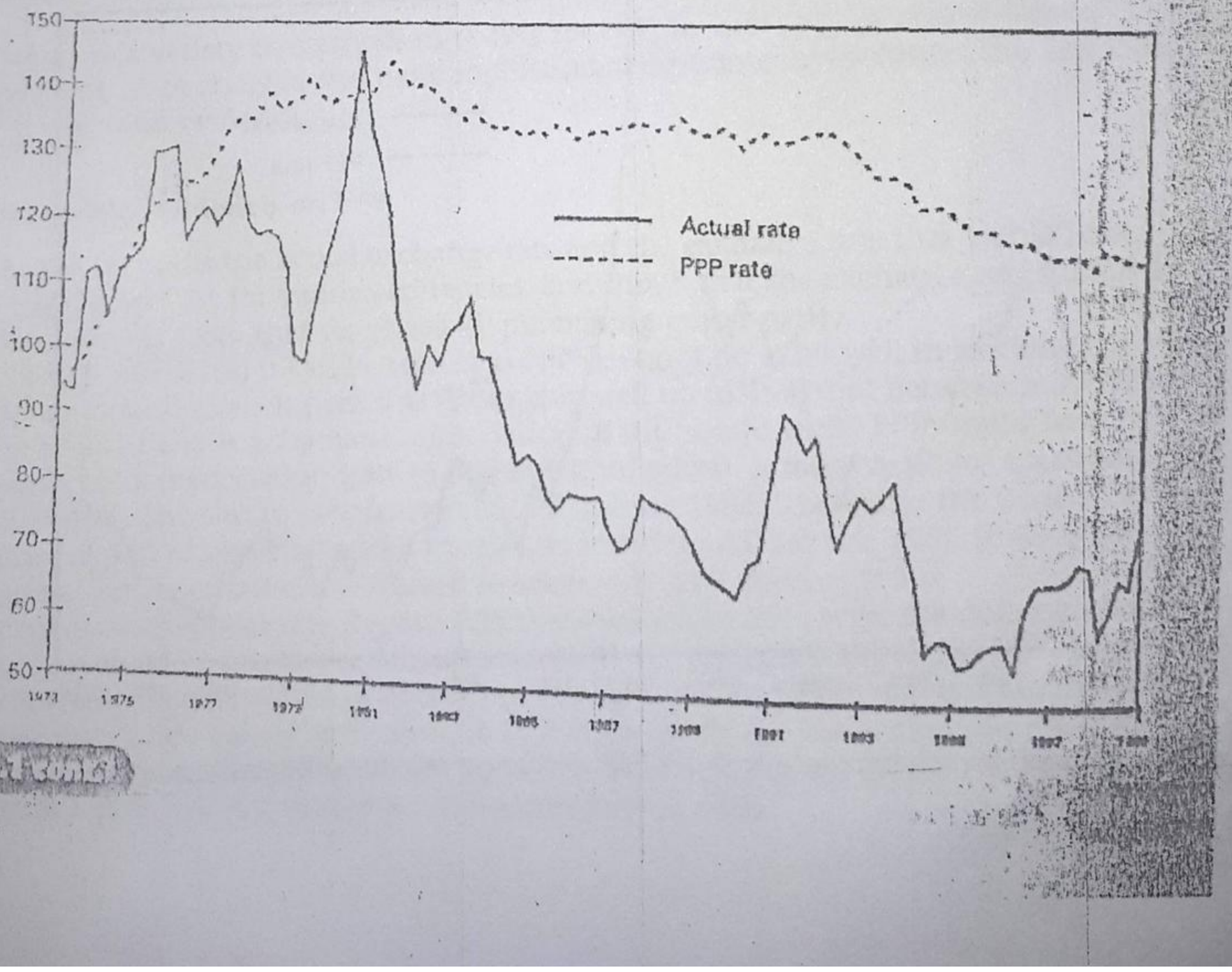
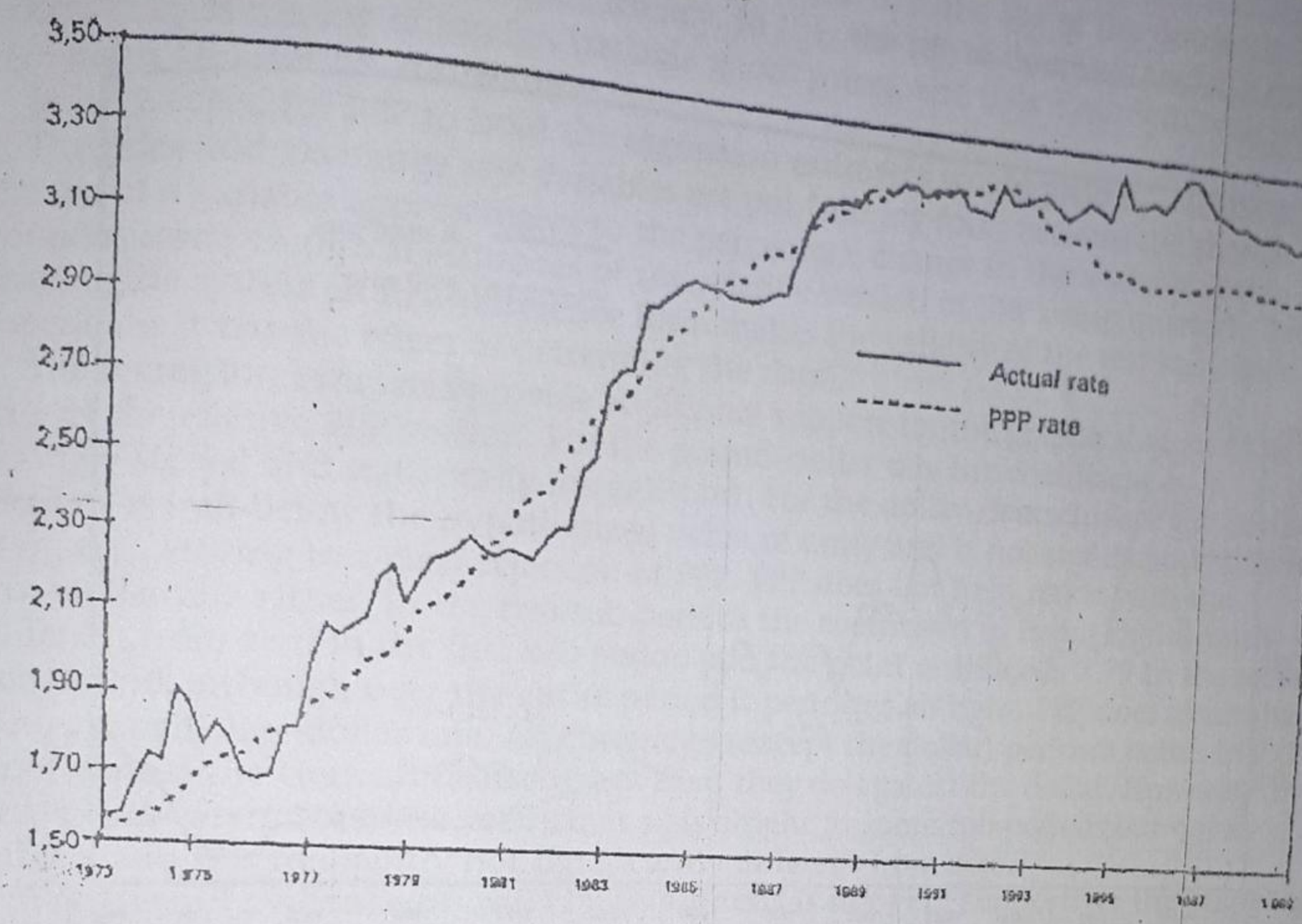


Figure 6.1 (cont.)

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(a) French franc–deutschmark rate and PPP rate



(f) Lira–deutschmark and PPP rate

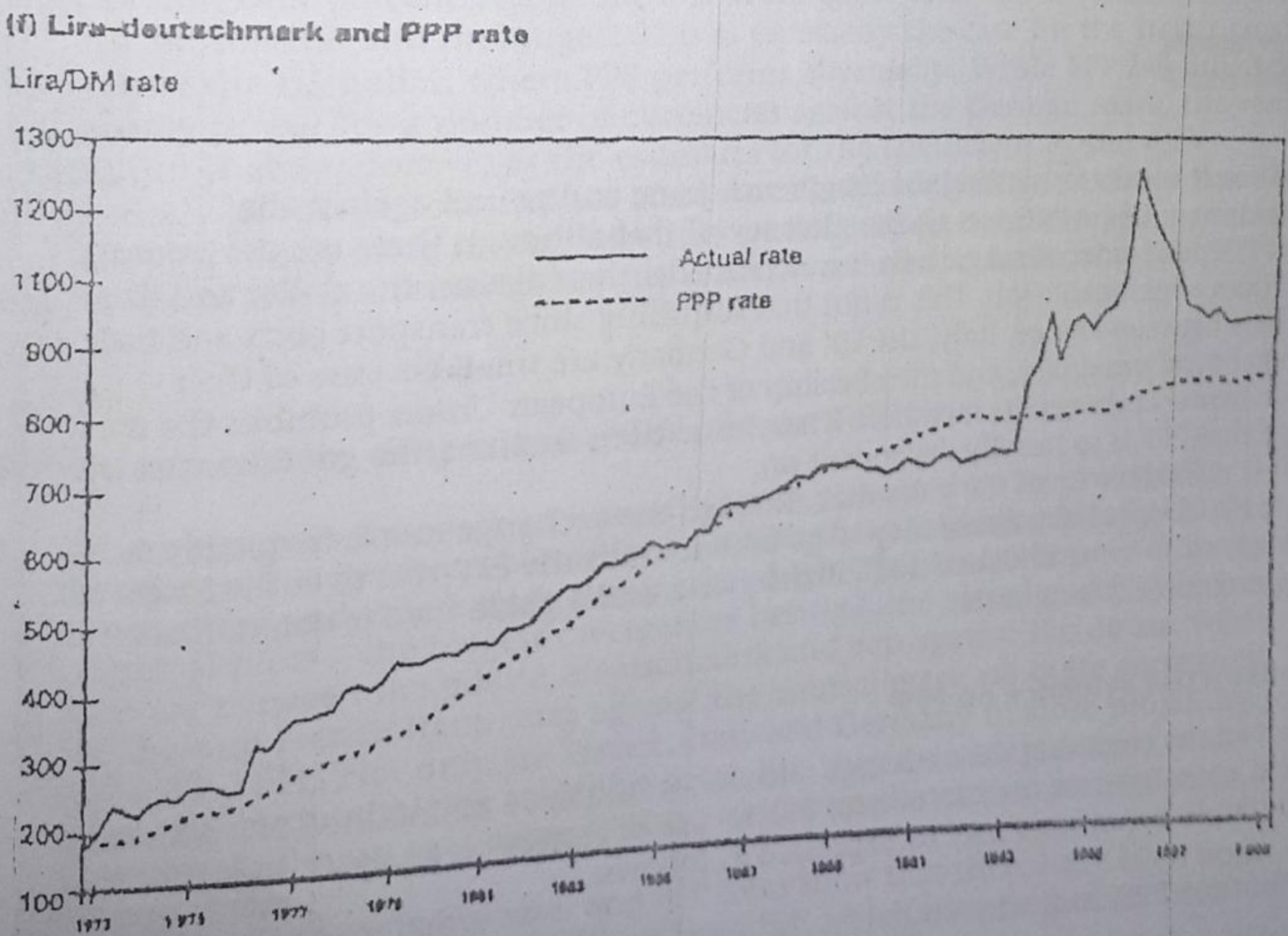
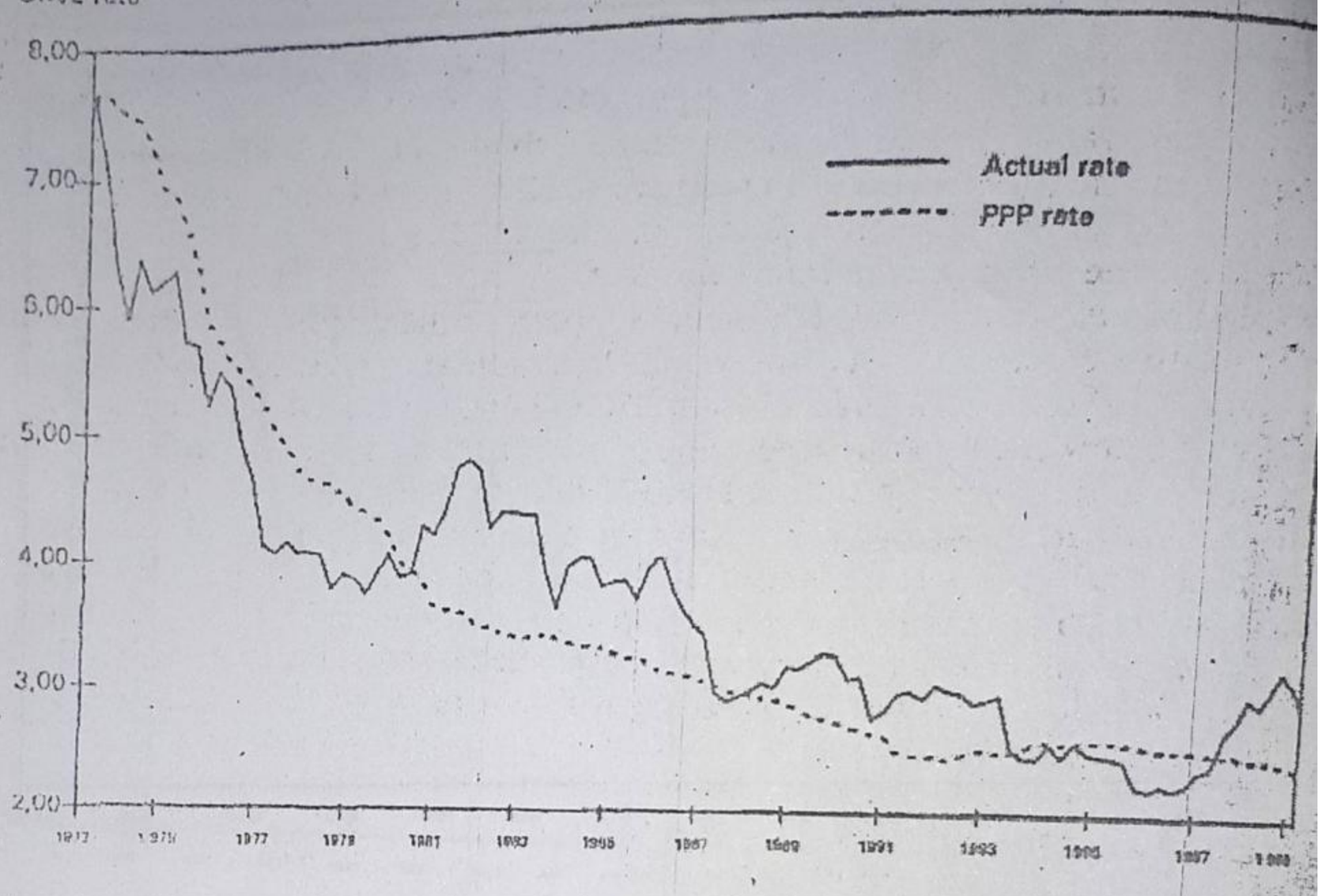


Figure 6.1 cont'd

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(g) Deutschmark-pound rate and PPP rate



When it comes to tracking the lira, French franc and pound against the deutschmark (Figures 6.1(e–g)) the plots reveal that although there are deviations from PPP, the order of magnitude is much smaller than against the dollar and that PPP does a reasonable job. This is not that surprising since transport costs and trade barriers between France, Italy, the UK and Germany are small because of their geographical proximity, and membership of the European Union prohibits the use of trade barriers between its members. These conditions facilitate the goods-market arbitrage that PPP is so heavily dependent on.

It is noticeable in all the plots that although the exchange rate is frequently far from PPP it does have a tendency to go back towards the PPP rate over the longer run. This provides some evidence that PPP may be a useful guide for the determination of the long run exchange rate.

Econometric evidence on PPP

Apart from comparing the exchange rate that would have maintained PPP with the actual exchange rate, we can test for PPP by use of regression analysis. In log form the two simple hypotheses may be expressed as follows:

$$\text{Absolute PPP: } \ln S_t = a_1 + a_2(\ln P_t - \ln P_{t*}) + u_t$$

$$\text{Relative PPP: } \Delta\ln S_t = a_1 + a_2(\Delta\ln P_t - \Delta\ln P_{t*}) + u_t$$

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where Δ is the one-period change in the variable, ln S_t is the log of the domestic exchange rate per unit of foreign currency, ln P_t is the log of domestic tradable goods prices, ln P_{t^*} is the log of foreign tradable goods prices, and u_t is a random error with zero mean and normal distribution.

In both cases, for PPP to hold the regression estimates would yield $a_1 = 0$ and $a_2 = 1$. The price and exchange rate variables are put into log form because the change in the log of a variable approximates to the percentage change in the variable. Table 6.1 contains some empirical estimates of the relative version of PPP using quarterly data (having the system in first-difference form makes the validity of the test somewhat superior as it has the effect of detrending the data).

The regression estimates provide additional support for the graphical analysis in the case of the relative PPP version. For the pound–dollar rate the coefficient a_2 is wrongly-signed and statistically insignificant. For the dollar–deutschmark the coefficient a_2 is well-below the hypothesized value of unity and is not significantly different from zero leading to a clear rejection of PPP. PPP does not help much with the yen–dollar rate either, in the two sub-periods the coefficient a_2 is not significantly different from zero in the first sub-period and the point estimate is 2.79 in the second sub-period, although over the entire period it performs all right. PPP does somewhat better for the lira–dollar rate. All currencies (except the dollar) perform better in PPP tests against the German deutschmark than they do against the dollar. However, the results are very mixed; the coefficient a_2 is alright in some sub-periods but not in others, and it is frequently not significantly different from zero.

The overall evidence on the PPP hypothesis is not very supportive; the graphical evidence and econometric results show that for some rates the deviations from PPP are both substantial and prolonged. This is especially the case for the major currencies against the US dollar, where PPP performs abysmally. While PPP has a better explanatory power for a number of currencies against the German mark, the results are still not that supportive, as the estimates for the coefficient a_2 can differ substantially in sub-periods and are often not significantly different from zero. In sum, there is much more to exchange rate determination than PPP. For an excellent review of the literature on PPP the reader is referred to Sarno and Taylor (2002).

6.7 Summary of the empirical evidence on PPP

1 Our results are very much in line with those presented by Frenkel (1981), that shows PPP performs better for countries that are geographically close to one another and where trade linkages are high. This is also borne out in the graphical plots – the biggest divergences between the actual and PPP exchange rates are between the pound, deutschmark and yen against the dollar; while the lira and French franc rates against the deutschmark are quite accurately tracked by PPP. Not only are France, Italy and Germany in close proximity to one another, minimizing transport costs, but they are also members of the European Union so that there exist no tariff impediments to restrict trade among them.

2 The plots of the exchange rates and PPP rates show that there have been both substantial and prolonged deviations from PPP which have frequently been reversed.

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TABLE 6.3 Relative purchasing power parity tests

Rate	Relative PPP Period	$\Delta \ln S_t = a_1 + a_2(\Delta \ln P_{t,} - \Delta \ln P_t) + u_t$ a_1	a_2	SE	DW
Pound/dollar	73Q1–81Q4	0.01 (0.56)	–0.17 (–0.46)	0.050	1.83
	81Q4–90Q3	0.00 (0.03)	–0.21 (–0.26)	0.059	1.85
	73Q1–90Q3	0.00 (0.41)	–0.01 (–0.04)	0.054	1.87
Deutschmark/dollar	73Q1–81Q4	0.00 (–0.25)	0.53 (0.89)	0.069	1.79
	81Q4–90Q3	–0.01 (–0.76)	0.48 (0.59)	0.062	1.96
	73Q1–90Q3	–0.01 (–0.80)	0.46 (1.00)	0.064	1.88
Yen/dollar	73Q1–81Q4	0.00 (–0.43)	0.82 (1.82)	0.051	1.94
	81Q4–90Q3	0.01 (0.93)	2.79 (3.67)	0.063	2.01
	73Q1–90Q3	0.00 (–0.26)	1.22* (3.15)	0.058	1.95
Lira/dollar	73Q1–81Q4	0.01 (0.74)	0.68* (2.25)	0.053	1.99
	81Q4–90Q3	–0.01 (–0.70)	0.78* (1.07)	0.055	1.90
	73Q1–90Q3	0.00 (–0.05)	0.73* (2.36)	0.055	1.98
French franc/deutschmark	73Q1–81Q4	0.00 (0.21)	0.77* (1.29)	0.034	1.89
	81Q4–90Q3	0.00 (0.59)	0.76* (1.35)	0.021	1.91
	73Q1–90Q3	0.00 (0.53)	0.71* (1.90)	0.027	2.00
Lira/deutschmark	73Q1–81Q4	0.01 (1.32)	0.51* (1.64)	0.054	1.80
	81Q4–90Q3	0.00 (0.56)	0.55* (2.39)	0.017	1.88
	73Q1–90Q3	0.00 (0.87)	0.68* (3.51)	0.040	1.79
Pound/deutschmark	73Q1–81Q4	0.01 (0.91)	0.16 (0.39)	0.057	1.95
	81Q4–90Q3	–0.01 (–0.71)	1.32* (2.63)	0.045	1.97
	73Q1–90Q3	0.01 (0.83)	0.40 (1.32)	0.051	1.96
Yen/deutschmark	73Q1–81Q4	0.00 (–0.25)	0.90* (1.84)	0.061	1.99
	81Q4–90Q3	0.00 (0.10)	1.18* (2.81)	0.039	1.97
	73Q1–90Q3	0.00 (–0.15)	0.93* (2.78)	0.050	1.99

Notes: Hypothesis is $a_2 = 1$. An asterisk by a variable indicates that it is both of the correct sign and statistically equal to its hypothesized value. The *t*-statistics are in parentheses.
Source: Author's own estimates.

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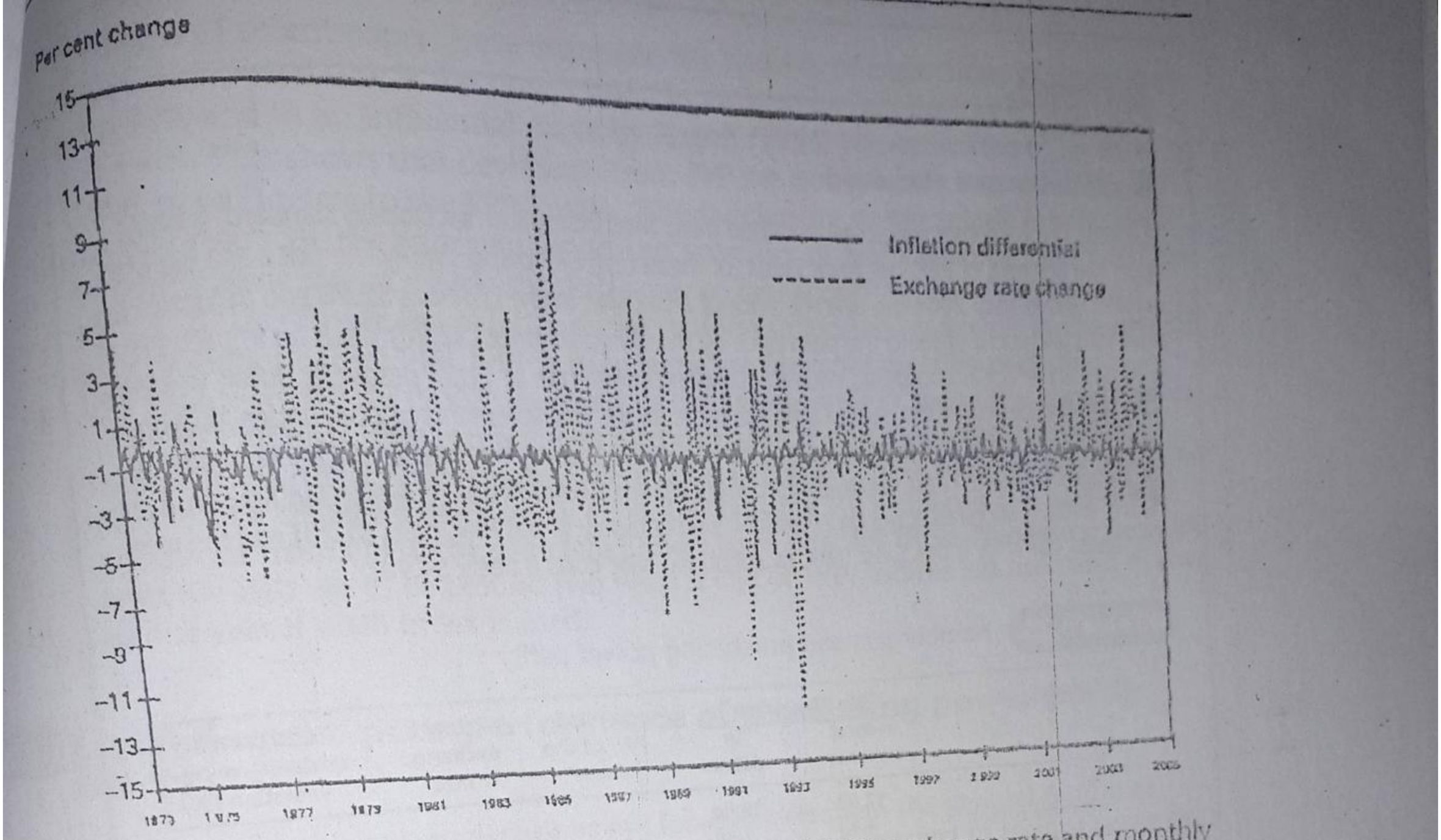


Figure 6.2 Monthly per cent changes in the dollar–pound ($/£) exchange rate and monthly inflation differentials (US–UK)

3 Exchange rates have been much more volatile than the corresponding national price levels (see Frenkel and Mussa (1980), and MacDonald (1988)). This again is contrary to the PPP hypothesis in which exchange rates are only supposed to be as volatile as prices. Figure 6.2 shows that the monthly volatility of the $/£ exchange rate is significantly higher than the difference in the monthly changes in the US and UK price indices.

4 Empirically, PPP holds better in the long run than in the short run. A study by Lothian (1997) using data from 22 OECD countries covering the period 1974–91 found that although there have been substantial and prolonged short-run deviations from relative PPP, it cannot be rejected over the longer run between three to six years. This long-run validity of PPP has also been supported by Sarno and Taylor (2002) using the latest econometric techniques.

5 The currencies of countries that had very high inflation rates relative to trading partners, such as Israel, Brazil and Argentina in the 1980s, experienced rapid depreciations reflecting their relatively high inflation rates. This suggests that PPP was the dominant force in determining their exchange rates during that period.

6 Overall, PPP holds better for traded goods than for non-traded goods, and this is confirmed in a study by Officer (1986). In addition, a striking and major empirical regularity is that the price of non-traded goods tends to be more expensive in rich countries than in poor countries once prices are converted into a common currency; we investigate this phenomenon further in section 6.10. This observation also has important implications about how we measure per capita GDP (see section 6.10).

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The hamburger standard

In 1986, *The Economist* magazine launched a Big Mac index; the 'McDonald standard' is based upon the concept of PPP, the price index for measuring PPP being simply the price of a Big Mac hamburger. In May 2004 the average price of a Big Mac in the USA was $2.90 and in Mexico the price was 24 pesos; dividing the peso price by the US dollar price ($2.90) yields an implied PPP of 8.28 pesos per dollar compared with an actual exchange rate of 11.57 pesos per dollar, yielding a 28% undervaluation of the peso. Table 6.2 presents the measurements of over/undervaluation of the dollar in terms of PPP against various other currencies using hamburger prices. According to the table, the second-most undervalued currency is the Chinese yuan (a Beijing Big Mac costs only $1.26), while Philippines is the most undervalued currency (a big Mac costs $1.23) and the most overvalued currency is the Swiss franc (a Zurich Big Mac costs a beefy $4.90).

Hamburgers and purchasing power parity

Country	Big Mac prices		Implied PPP of the dollar*	Actual $ exchange rate	Local currency under(–)/over(+) valuation (%)
	In local currency	In dollars			
USA	$2.90	$2.90	–	–	–
Argentina (peso)	4.35	1.48	1.50	2.96	–49
Australia (A$)	3.25	2.27	1.12	1.43	–22
Brazil (real)	5.40	1.70	1.86	3.18	–41
Britain (£)	1.88	3.37	1.54**	1.79	+16
Canada (C$)	3.20	2.33	1.10	1.37	–20
Chile (peso)	1400	2.18	483	642	–25
China (yuan)	10.40	1.26	3.59	8.25	–56
Czech Re (CKr)	56.55	2.13	19.50	26.55	–27
Denmark (DKr)	27.75	4.46	9.57	6.22	+54
Egypt	10.00	1.62	3.45	6.17	–44
Euro Area (€)	2.74	3.28	1.06***	1.20	+13
HongKong (HK$)	12.00	1.54	4.14	7.79	–47
Hungary (forint)	531	2.52	183	210	–13
Indonesia (rupiah)	16,100	1.77	5,552	9,096	–39
Japan (yen)	262	2.33	90.3	112.4	–20
Malaysia (M$)	5.05	1.33	1.74	3.79	–54
Mexico (peso)	24.00	2.08	8.28	11.54	–28
New Zealand (NZ$)	4.35	2.65	1.50	1.64	–9
Peru	9.00	2.57	3.10	3.50	–11
Philippines	69.00	1.23	23.8	56.1	–58
Poland (zloty)	6.29	1.63	2.17	3.86	–44
Russia (roubie)	42.05	1.45	14.5	29.0	–50
Singapore (S$)	3.30	1.92	1.14	1.72	–34
South Africa (rand)	12.40	1.86	4.28	6.67	–36
S.Korea (won)	3,200	2.72	1,103	1,176	–6
Sweden (SKr)	29.90	3.94	10.3	7.59	+36
Switzerland (SFr)	6.30	4.90	2.17	1.29	+68
Taiwan (NT$)	75.10	2.24	25.9	33.5	–23
Thailand (baht)	58.90	1.45	20.3	40.6	–50
Venezuela	4,400	1.48	1,517	2,973	–49

Local currency prices and actual exchange rates are inferred by the author from data presented in *The Economist*. 'Euro Area' is a weighted average price based on the price in the 12 Eurozone countries. * PPP is the local price divided by the price in the United States, ** dollars per pound, *** dollars per euro.
Source: *The Economist*, 27 May 2004.

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A number of recent papers have estimated the half-life of corrections of deviations from PPP. The half-life is the time it takes in years to correct 50% of a deviation from PPP, and in an influential paper by Rogoff (1996) was estimated to be three to five years. This shows that deviations from PPP are not speedily corrected. Rogoff (1996, p. 647) refers to the PPP Puzzle, "The purchasing power puzzle is this. How can one reconcile the enormous short run volatility of real exchange rates with the extremely slow rate at which shocks appear to damp out?' However, Sarno and Taylor (2002) dispute the validity of the half-lives reported by Rogoff in the case of large deviations from PPP, which they argue are far more speedily corrected. In a recent study, Xu (2003) emphasizes the importance of using the correct price index in PPP tests. PPP is much less likely to hold if one uses consumer price indices (CPI), than if wholesale price indices (WPI) are used; significantly better is the use of a tradables price index (TPI). In his study Xu estimates the half-life of deviations from PPP to be somewhat lower than Rogoff's three–five-years; he calculates the half life to be around two years if CPI or WPI indices are used and as low as one year if a TPI index is used.

8

Explaining the poor performance of purchasing power parity theory

There have been many explanations put forward to explain the general failure of exchange rates to adjust in line with PPP theory, and we proceed to look at some of the most important.

Statistical problems

We have seen that PPP theory is based upon the concept of comparing identical baskets of goods in two economies. An important problem facing researchers in this respect is that different countries usually attach different weighs to various categories of goods and services when constructing their price indices, which means it is difficult to compare 'like with like' when testing for PPP. This factor is probably very significant when testing for PPP between developed and developing economies which have vastly different consumption patterns. People in developing countries usually spend a high proportion of their income on basics such as food and clothing, while these take up a much smaller proportion of people's expenditure in developed economies.

Differing consumption baskets are not of such significance when comparing most industrialized economies since consumers have fairly similar consumption baskets in these economies. Even between developed economies, however, there is a problem posed by the differing quality of goods consumed. Although British and German consumers both spend roughly the same proportion of their incomes on cars, the Germans tend to drive German makes like BMW while the British tend to drive Fords and Japanese cars. We do not necessarily expect PPP to hold in terms of cars between the two countries because once again we are not comparing like with like.

Transport costs and trade impediments

Studies such as Frenkel (1981), which note that PPP holds better when the countries concerned are geographically close and trade linkages are high, can partly be

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explained by transport costs and the existence of other trade impediments such as tariffs. If a bundle of goods costs £100 in the UK and $200 in the USA, PPP would suggest an exchange rate of £0.50/$1. If transport costs are £20 then the exchange rate could lie anywhere between £0.40/$1 and £0.60/$1 without bringing arbitrage forces into play. Nonetheless, since transport costs and trade barriers do not change dramatically over time they are not sufficient explanations for the failure of the relative versions of PPP.

Imperfect competition

One of the notions underlying purchasing power parity is that there is sufficient international competition to prevent major departures of the price of a good in one country exceeding that in another. However, it is clear that there are considerable variations in the degree of competition internationally, and these differences mean that multinational corporations can often get away with charging different prices in different countries. In fact, the conditions necessary for successful price discrimination, namely differences in the willingness to pay of different sets of consumers, the ability to prevent resale from the low-cost to high-cost market and some degree of monopoly power, are for the most part more likely to hold between rather than within countries.

Differences between capital and goods markets

Purchasing power parity is based upon the concept of goods arbitrage and has nothing to say about the role of capital movements. In a classic paper which we shall be looking at in more detail in Chapter 7, Rudiger Dornbusch (1976) hypothesized that in a world where capital markets are highly integrated and goods markets exhibit slow price adjustment, there can be substantial prolonged deviations of the exchange rate from PPP. The basic idea is that in the short run goods prices in both the home and foreign economies can be considered as fixed, while the exchange rate adjusts quickly to new information and changes in economic policy. This being the case, exchange rate changes represent deviations from PPP which can be quite substantial and prolonged.

Productivity differentials

As mentioned, one striking empirical observation that is well-documented is that when prices of similar baskets of both traded and non-traded goods are converted into a common currency, the aggregate price indices tend to be higher in rich countries than in poor countries. In other words, a dollar buys more goods in say Mexico than in the United States. Further, evidence shows that prices of tradable goods are nowhere as dissimilar internationally as those of non-traded goods. Consequently the overall higher price index in rich countries is mainly due to the fact that non-tradable goods prices are higher in developed than developing countries.

An explanation for the lower relative price of non-tradables in poor countries has been put forward separately by Bela Balassa (1964) and Paul Samuelson (1964), and is worth considering in detail. As we shall see, the model argues that the lower price of goods in the non-traded sector in developing countries is mainly due to lower labour productivity in the tradables sector compared to developed countries. Or, equivalently,

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the higher price of non-traded goods in a developed nation (compared to non-traded goods in the developing country) is mainly due to higher labour productivity in the traded sector compared to labour productivity in the tradables sector of developing countries.

6.9 The Balassa–Samuelson model

Balassa (1964) and Samuelson (1964) argue that labour productivity in rich countries is higher than labour productivity in poor countries. Furthermore, this productivity differential occurs predominately in the tradables rather than the non-tradables sector. A Mexican barber tends to be as efficient as his American counterpart (as measured by haircuts per day), but a Mexican car worker is less efficient than his American counterpart (as measured by cars produced per day). Wages are assumed to be the same in the tradables and non-tradables sectors within each economy and positively related to productivity. Prices are determined positively by wages and inversely by productivity. These assumptions lead to the following set of relationships:

In the poor country:

$$P_N = W_N/Q_N \quad \text{and} \quad P_T = W_T/Q_T \tag{6.9}$$

In the rich economy:

$$P_{N^*} = W_{N^*}/Q_{N^*} \qquad P_{T^*} = W_{T^*}/Q_{T^*} \tag{6.10}$$

where P_N represents non-tradables prices, P_T represents traded goods prices, Q_N is output per worker in the tradables sector, Q_T is output per worker in the non-tradables sector, and an asterisk denotes a high income (high productivity) economy.

Wage rates are the same in both the industries in each of the two economies:

$$W_N = W_T \quad \text{and} \quad W_{N^*} = W_{T^*} \tag{6.11}$$

Productivity is higher in the rich economy's tradables sector than in the poor country's, but in the non-tradables sector productivity in the rich and poor countries are the same. This means that:

$$Q_{T^*} > Q_T \quad \text{and} \quad Q_{N^*} = Q_N \tag{6.12}$$

Finally, PPP is assumed to hold only for traded goods:

$$S = P_T/P_{T^*} \tag{6.13}$$

The price ratio of traded to non-traded goods in each country is given by:

$$\frac{P_N}{P_T} = \pi \tag{6.14}$$

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$$\frac{P_{N^*}}{P_{T^*}} = \pi^* \tag{6.15}$$

Since there is higher productivity in the rich economy's tradables sector, then the relative price of non-tradables to tradables will be higher in the rich country making $\pi^* > \pi$.

Rewriting equations (6.14) and (6.15) as:

$$\frac{P_N}{P_T} = \pi \tag{6.16}$$

and

$$\frac{SP_{N^*}}{SP_{T^*}} = \pi^* \tag{6.17}$$

Since PPP holds for the tradables sector, making $P_T = S\,P_{T^*}$, then the denominator of equations (6.16) and (6.17) are the same. Consequently, since $\pi^* > \pi$, the price of non-tradables in the rich economy exceeds the price of non-tradables in the developing country, that is, $S\,P_{N^*} > P_N$; that is, PPP does not work for non-traded goods. The reason is that low wages in the developing country due to low productivity in its traded sector also lead to a relatively low price for its non-traded goods even though its productivity in this sector is the same as in developed countries; while high productivity in the rich country's tradables sector leads to high wages in its non-tradables sector, even though it is no more efficient than the poor country in that sector. Consequently, when we use the exchange rate to examine non-tradable goods prices we find that they are higher in developed countries than developing countries.

The Balassa–Samuelson model is helpful in explaining why it is that rich countries tend to have overall high price indices and poor countries low price indices when aggregate baskets of traded and non-traded goods are converted into a common currency such as the US dollar. Furthermore, it helps explain why the ratio of non-traded to traded prices tends to be higher in developed economies than developing countries.

It is clear that PPP based on tradable goods alone will undervalue the purchasing power and living standards of people in poor countries because they can purchase more non-traded goods per dollar than can people in rich countries, even if PPP holds for traded goods (see section 6.10). This means that when comparing per capita income levels between countries such as Mexico and the United States in dollar terms this should not be done at the PPP rate for tradables goods; rather, the proportion of Mexican expenditure on non-tradables should be valued at US non-tradables prices thereby raising the value of Mexican incomes in US dollar terms.

The productivity differential theory may also have some application in explaining divergences from PPP in terms of aggregate price indices between developed countries. For instance, since the Second World War Japan has consistently had higher productivity in its tradables sector than the United States. According to the Balassa–Samuelson model, this means that the real value of the yen should appreciate against the dollar as the higher productivity of Japanese tradables workers leads to a

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fall in Japanese traded goods prices relative to US traded goods prices, and therefore an appreciation of the yen to maintain PPP for tradable goods. Richard Marston (1986) found evidence that in the period 1973–83, Japanese productivity growth in its tradables sector outstripped US tradables productivity. During that period the yen appreciated by some 9% in real terms against the dollar.

Although the Balassa–Samuelson model helps explain why PPP does not necessarily hold in terms of aggregate price indices, it is only a partial explanation. By assumption the theory cannot explain the failure of PPP to hold for traded goods!

6.10 Per capita income levels, the relative sizes of economies and the importance of PPP estimates

The World Bank keeps a very keen eye on the per capita income levels of developing countries, since the poorest countries in the world are often deemed to be the countries that policy-makers wish to direct aid towards and provide access to cheap finance for developmental purposes. The fact that PPP tends not to hold, especially between developed and developing nations, is of considerable importance in this respect. It is well-documented that, in particular, the price of non-tradable goods and services in developing countries is significantly lower in developing countries than in developed countries. This means that $100 will tend to buy far more haircuts, restaurant meals, hotel time and many other goods and services when transferred into the local currency of a developing country than in a developed nation like the United States, the United Kingdom or Japan, simply because PPP is not holding. In particular, the exchange rates of developing countries tend to be noticeably undervalued in terms of purchasing power for goods and services. Consider the case of the Chinese renminbi versus the US dollar; the official exchange rate in 2004 was 8.6 renminbi per US dollar. Imagine, however, that when we change $100 into renminbi to obtain 860 renminbi we are then able to buy five times more goods and services in China than in the USA. The appropriate PPP exchange rate in terms of goods and services would then be 8.6/5 = 1.72 renminbi per dollar. If the exchange rate was at the calculated PPP rate then $100 would convert to 172 renminbi which would purchase the same bundle of goods and services in China as $100 buys in the USA. Think now of the implications for calculations of GDP per capita. If the United States GDP per capita was $35,000 while the Chinese per capita income was 8,600 renminbi, then using the market exchange rate the Chinese income per capita is only $1,000 which is a mere 1/35th of that of the United States. However, if $1,000 buys five times more goods and services in China, then we need to multiply the Chinese income by five to get a PPP per capita income which is $5,000 (which is the same as dividing the Chinese GDP per capita of 8,600 renminbi by the PPP rate of 1.72). This would suggest that Chinese income levels are 'only' 1/7th of those of the United States. The GDP per capita based on PPP is therefore a more reliable guide (albeit highly imperfect!) to relative living standards in the two countries than using the market exchange rate which does not reflect purchasing power parity.

The previous example calculation is by no means far from reality. In the case of the United States and China, as Table 6.3 shows gross national income (GDP) per capita according to the World Bank using the both the Atlas method and the PPP method. The Atlas method mainly uses an average of market exchange rates over the past three years to calculate income per capita in terms of US dollars, making some allowance for recent inflation (this methodology helps to iron out sudden changes in relative figures

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STARTER GDP per capita using the Atlas and PPP methods

	Atlas method (US dollars)	PPP method (PPP dollars)
Luxembourg	43,940	54,430
Norway	43,450	37,300
Switzerland	39,880	32,030
United States	37,610	37,300
Japan	34,510	26,620
Denmark	33,650	31,210
Sweden	28,840	26,620
United Kingdom	28,350	27,650
Finland	27,020	27,100
Ireland	26,960	30,450
Netherlands	26,310	28,600
Belgium	25,820	28,930
Germany	25,250	27,460
France	24,770	27,460
Canada	23,930	29,740
Australia	21,650	28,920
Italy	21,560	26,760
Singapore	21,230	24,180
Spain	16,990	22,020
Greece	13,720	19,920
Slovenia	11,830	19,240
Czech Republic	6,740	15,650
Hungary	6,330	13,780
Mexico	6,230	8,950
Poland	4,960	11,450
Malaysia	3,780	8,940
Argentina	3,650	10,920
Venezuela	3,490	4,740
Turkey	2,790	6,690
South Africa	2,780	10,270
Brazil	2,710	7,480
Russia	2,610	8,920
Egypt	1,390	3,940
China	1,100	4,990
India	530	2,880
Kenya	390	960
Mozambique	210	1,070

Source: World Bank, figures are for 2003.

due merely to large year-to-year exchange rate changes compared to using purely current market exchange rates), and these are compared to the estimates of GDP per capita using PPP estimates in the second column.

Not only do PPP adjustments make a significant difference to GDP per capita estimates, they also make a big difference to the importance of the relative size of different economies. As Table 6.4 shows, the largest economy in 2003 using market exchange

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TABLE 6.1 The sizes of different economies as measured by their GDPs

Rank using market rates		$millions	Rank using PPP rates		$millions PPP
1	United States	10,881,609	1	United States	10,871,095
2	Japan	4,326,444	2	China	6,435,838
3	Germany	2,400,655	3	Japan	3,582,515
4	United Kingdom	1,794,858	4	India	3,096,239
5	France	1,747,973	5	Germany	2,279,134
6	Italy	1,465,895	6	France	1,632,119
7	China	1,409,852	7	United Kingdom	1,606,853
8	Spain	836,100	8	Italy	1,559,321
9	Canada	834,390	9	Brazil	1,371,655
10	Mexico	626,080	10	Russia	1,318,827
11	Korea (South)	605,331	11	Canada	963,550
12	India	598,966	12	Mexico	934,553
13	Australia	518,382	13	Spain	915,072
14	Netherlands	511,556	14	Korea	858,028
15	Brazil	492,338	15	Indonesia	721,583
16	Russia	433,491	16	Australia	579,662
17	Switzerland	309,465	17	Turkey	477,256
18	Belgium	302,217	18	Netherlands	476,912
19	Sweden	300,795	19	South Africa	475,215
20	Austria	251,456	20	Iran	474,383

Source: World Bank, figures are for 2003.

rates was the United States with Japan second and China seventh. If, however, we measure the relative sizes of economies based upon PPP exchange rate estimates then China becomes the second largest economy in the world, with India jumping from twelfth to fourth and Brazil jumping from fifteenth to ninth!

6.11 Conclusions

At the time of the adoption of floating exchange rates it was widely believed that they would adjust in line with changes in national price levels as predicted by PPP theory. However, the experience with floating rates has shown that there can be substantial and prolonged deviations of exchange rates from PPP. A clear conclusion is that in the short to medium term, international goods arbitrage is nowhere near as powerful as proponents of PPP had presupposed.

There are many possible explanations for these deviations from PPP. Among the strongest candidates is that the theory relies too heavily on goods arbitrage and has no role for the international capital movements which have grown enormously in scale since the end of the Second World War. Such capital movements are heavily influenced by prospective returns and therefore agents' expectations about the future. As expectations change, then so will exchange rates regardless of whether goods prices are changing. In the following chapters we shall be looking at some of the most recent theories that attempt to take into account the implications of such capital movements.

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Nonetheless, the fact that PPP does not hold very well in the short to medium term does not mean that it has no role to play in exchange rate determination. Over- or undervaluation of currencies in relation to PPP induce changes in current-account positions which will eventually lead to exchange rate changes. It is the case that deviations from PPP do have a habit of reversing themselves over the longer run. Furthermore, although exchange rates may diverge substantially from PPP, if the break in this link becomes too large the forces of goods arbitrage do start to come into play and move the exchange rate towards its PPP value. More recent evidence shows that PPP may be a useful guide to the long-run exchange rate and should not therefore be abandoned altogether.

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The Balance of Payments Accounts

In addition to national income accounts, government economists and statisticians also keep balance of payments accounts, a detailed record of the composition of the current account balance and of the many transactions that finance it.[10] Balance of payments figures are of great interest to the general public, as indicated by the attention that various news media pay to them. But press reports sometimes confuse different measures of international payments flows. Should we be alarmed or cheered by a *Wall Street Journal* headline proclaiming, "U.S. Chalks Up Record Balance of Payments Deficit"? A thorough understanding of balance of payments accounting will help us evaluate the implications of a country's international transactions.

A country's balance of payments accounts keep track of both its payments to and its receipts from foreigners. Any transaction resulting in a receipt from foreigners is entered in the balance of payments accounts as a *credit*. Any transaction resulting in a payment to foreigners is entered as a *debit*. Three types of international transaction are recorded in the balance of payments:

1. Transactions that arise from the export or import of goods or services and therefore enter directly into the current account. When a French consumer imports American blue jeans, for example, the transaction enters the U.S. balance of payments accounts as a credit on the current account.
2. Transactions that arise from the purchase or sale of financial assets. An **asset** is any one of the forms in which wealth can be held, such as money, stocks, factories, or government debt. The **financial account** of the balance of payments records all international purchases or sales of financial assets. When an American company buys a French factory, the transaction enters the U.S. balance of payments as a debit in the financial account. It enters as a debit because the transaction requires a payment from the United States to foreigners. Correspondingly, a U.S. sale of assets to foreigners enters the U.S. financial account as a credit. The difference between a country's purchases and sales of foreign assets is called its *financial account balance*, or its *net financial flows*.
3. Certain other activities resulting in transfers of wealth between countries are recorded in the **capital account.** These international asset movements—which are generally very small for the United States—differ from those recorded in the financial account. For the most part they result from nonmarket activities or represent the acquisition or disposal of nonproduced, nonfinancial, and possibly intangible assets (such as copyrights and trademarks). For example, if the U.S. government forgives $1 billion in debt owed to it by the government of Pakistan, U.S. wealth declines by $1 billion and a $1 billion debit is recorded in the U.S. capital account.

You will find the complexities of the balance of payments accounts less confusing if you keep in mind the following simple rule of double-entry bookkeeping: *Every international transaction automatically enters the balance of payments twice, once as a credit and*

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[10]The U.S. Bureau of Economic Analysis (BEA) has changed its balance of payments presentation to conform to prevailing international standards, so our discussion in this chapter differs in some respects from that in editions one through eight of this book. We follow the new methodology described by Kristy L. Howell and Robert E. Yuskavage, "Modernizing and Enhancing BEA's International Economic Accounts: Recent Progress and Future Directions," *Survey of Current Business* (May 2010), pp. 6–20. The BEA completed the full transition to the new system in June 2014. For an update, see Jeffrey R. Bogen, Mai-Chi Hoang, Kristy L. Howell, and Erin M. Whitaker, "Comprehensive Restructuring and Annual Revision of the U.S. International Transactions Accounts," *Survey of Current Business* (July 2014), pp. 1–24.

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once as a debit. This principle of balance of payments accounting holds true because every transaction has two sides: If you buy something from a foreigner, you must pay him in some way, and the foreigner must then somehow spend or store your payment.

Examples of Paired Transactions

Some examples will show how the principle of double-entry bookkeeping operates in practice.

1. Imagine you buy an ink-jet fax machine from the Italian company Olivetti and pay for your purchase with a $1,000 check. Your payment to buy a good (the fax machine) from a foreign resident enters the U.S. current account as a debit. But where is the offsetting balance of payments credit? Olivetti's U.S. salesperson must do something with your check—let's say he deposits it in Olivetti's account at Citibank in New York. In this case, Olivetti has purchased, and Citibank has sold, a U.S. asset—a bank deposit worth $1,000—and the transaction shows up as a $1,000 credit in the U.S. financial account. The transaction creates the following two offsetting bookkeeping entries in the U.S. balance of payments:

	Credit	Debit
Fax machine purchase (Current account, U.S. good import)		$1,000
Sale of bank deposit by Citibank (Financial account, U.S. asset sale)	$1,000	

2. As another example, suppose that during your travels in France, you pay $200 for a fine dinner at the Restaurant de l'Escargot d'Or. Lacking cash, you place the charge on your Visa credit card. Your payment, which is a tourist expenditure, will be counted as a service import for the United States, and therefore as a current account debit. Where is the offsetting credit? Your signature on the Visa slip entitles the restaurant to receive $200 (actually, its local currency equivalent) from First Card, the company that issued your Visa card. It is therefore an asset, a claim on a future payment from First Card. So when you pay for your meal abroad with your credit card, you are selling an asset to France and generating a $200 credit in the U.S. financial account. The pattern of offsetting debits and credits in this case is:

	Credit	Debit
Meal purchase (Current account, U.S. service import)		$200
Sale of claim on First Card (Financial account, U.S. asset sale)	$200	

3. Imagine next that your Uncle Sid from Los Angeles buys a newly issued share of stock in the U.K. oil giant British Petroleum (BP). He places his order with his U.S. stockbroker, Go-for-Broke, Inc., paying $95 with funds from his Go-for-Broke money market account. BP, in turn, deposits the $95 Sid has paid into its own U.S. bank account at Second Bank of Chicago. Uncle Sid's acquisition of the stock creates a $95 debit in the U.S. financial account (he has purchased an

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asset from a foreign resident, BP), while BP's $95 deposit at its Chicago bank is the offsetting financial account credit (BP has expanded its U.S. asset holdings). The mirror-image effects on the U.S. balance of payments therefore both appear in the financial account:

	Credit	Debit
Uncle Sid's purchase of a share of BP (Financial account, U.S. asset purchase)		$95
BP's deposit of Uncle Sid's payment at Second Bank of Chicago (Financial account, U.S. asset sale)	$95	

4. Finally, let's consider how the U.S. balance of payments accounts are affected when U.S. banks forgive (that is, announce that they will simply forget about) $5,000 in debt owed to them by the government of the imaginary country of Bygonia. In this case, the United States makes a $5,000 capital transfer to Bygonia, which appears as a $5,000 debit entry in the capital account. The associated credit is in the financial account, in the form of a $5,000 reduction in U.S. assets held abroad (a negative "acquisition" of foreign assets, and therefore a balance of payments credit):

	Credit	Debit
U.S. banks' debt forgiveness (Capital account, U.S. transfer payment)		$5,000
Reduction in banks' claims on Bygonia (Financial account, U.S. asset sale)	$5,000	

These examples show that many circumstances can affect the way a transaction generates its offsetting balance of payments entry. We can never predict with certainty where the flip side of a particular transaction will show up, but we can be sure that it will show up somewhere.

The Fundamental Balance of Payments Identity

Because any international transaction automatically gives rise to offsetting credit and debit entries in the balance of payments, the sum of the current account balance and the capital account balance automatically equals the financial account balance:

$$\text{Current account} + \text{capital account} = \text{Financial account}. \tag{2-3}$$

In examples 1, 2, and 4 previously, current or capital account entries have offsetting counterparts in the financial account, while in example 3, two financial account entries offset each other.

You can understand this identity another way. Recall the relationship linking the current account to international lending and borrowing. Because the sum of the current and capital accounts is the total change in a country's net foreign assets (including, through the capital account, nonmarket asset transfers), that sum necessarily equals the difference between a country's purchases of assets from foreigners and its sales of assets to them—that is, the financial account balance (also called net financial flows).

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We now turn to a more detailed description of the balance of payments accounts, using as an example the U.S. accounts for 2015.

The Current Account, Once Again

As you have learned, the current account balance measures a country's net exports of goods and services. Table 2-2 shows that U.S. exports (on the credit side) were $3,044.08 billion in 2015, while U.S. imports (on the debit side) were $3,362.06 billion.

The balance of payments accounts divide exports and imports into three finer categories. The first is *goods* trade, that is, exports or imports of merchandise. The second category, *services*, includes items such as payments for legal assistance, tourists' expenditures, and shipping fees. The final category, *income*, is made up mostly of international interest and dividend payments and the earnings of domestically owned firms

Pearson MyLab Economics Real-time data

TABLE 2-2 U.S. Balance of Payments Accounts for 2015 (billions of dollars)

Current Account	
(1) Exports	**3,044.08**
Of which:	
Goods	1,510.30
Services	750.86
Income receipts (primary income)	782.92
(2) Imports	**3,362.06**
Of which:	
Goods	2,272.87
Services	488.66
Income payments (primary income)	600.53
(3) Net unilateral transfers (secondary income)	**−144.99**
Balance on current account [(1) − (2) + (3)]	**−462.97**
Capital Account	
(4)	**−0.04**
Financial Account	
(5) Net U.S. acquisition of financial assets, excluding financial derivatives	**225.40**
Of which:	
Official reserve assets	
Other assets	−6.29
(6) Net U.S. incurrence of liabilities, excluding financial derivatives	231.69
Of which:	**395.23**
Official reserve assets	
Other assets	−98.10
(7) Financial derivatives, net	493.33
Net financial flows	**−25.39**
[(5) − (6) + (7)]	**−195.23**
Statistical Discrepancy	
[Net financial flows less sum of current and capital accounts]	**267.78**

Source: U.S. Department of Commerce, Bureau of Economic Analysis, June 16, 2016, release. Totals may differ from sums because of rounding.

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operating abroad. If you own a share of a German firm's stock and receive a dividend payment of $5, that payment shows up in the accounts as a U.S. investment income receipt of $5. Wages that workers earn abroad can also enter the income account.

We include income on foreign investments in the current account because that income really is compensation for the *services* provided by foreign investments. This idea, as we saw earlier, is behind the distinction between GNP and GDP. When a U.S. corporation builds a plant in Canada, for instance, the productive services the plant generates are viewed as a service export from the United States to Canada equal in value to the profits the plant yields for its American owner. To be consistent, we must be sure to include these profits in American GNP and not in Canadian GNP. Remember, the definition of GNP refers to goods and services generated by a country's factors of production, but it does *not* specify that those factors must work within the borders of the country that owns them. The earnings of capital and labor working abroad are referred to as "primary income."

Before calculating the current account, we must include one additional type of international transaction that we have largely ignored until now. In discussing the relationship between GNP and national income, we defined unilateral transfers between countries as international gifts, that is, payments that do not correspond to the purchase of any good, service, or asset. Such payments are referred to as "secondary income." Net unilateral transfers are considered part of the current account as well as a part of national income, and the identity $Y = C + I + G + CA$ holds exactly if Y is interpreted as GNP *plus* net transfers. In 2015, the U.S. balance of unilateral transfers was −$144.99 billion.

The table shows a 2015 current account balance of $3,044.08 billion − $3,362.06 billion − $144.99 billion = −$462.97 billion, a deficit.

The negative sign means that current payments to foreigners exceeded current receipts and that U.S. residents used more output than they produced. Since these current account transactions were paid for in some way, we know that this $462.97 billion net debit entry must be offset by a net $462.97 billion credit elsewhere in the balance of payments.

The Capital Account

The capital account entry in Table 2-2 shows that in 2015, the United States payed about $40 million in net capital transfers. The net balance of −$40 million is a balance of payments debit. After we add it to the payments deficit implied by the current account, we find that the United States' need to cover its excess payments to foreigners is increased very slightly, from $462.97 billion to $463.01 billion. Because an excess of national spending over income must be covered by net borrowing from foreigners, this negative current plus capital account balance must be matched by an equal negative balance of net financial flows, representing the net liabilities the United States incurred to foreigners in 2015 in order to fund its deficit.

The Financial Account

While the current account is the difference between sales of goods and services to foreigners and purchases of goods and services from them, the financial account measures the difference between acquisitions of assets from foreigners and the buildup of liabilities to them. When the United States borrows $1 from foreigners, it is selling them an asset—a promise that they will be repaid $1, with interest, in the future. Likewise, when the United States lends abroad, it acquires an asset: the right to claim future repayment from foreigners.

To cover its 2015 current plus capital account deficit of $463.01 billion, the United States needed to borrow from foreigners (or otherwise sell assets to them) in the net

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amount of $463.01 billion. We can look again at Table 2-2 to see exactly how this net sale of assets to foreigners came about.

The table records separately U.S. acquisitions of foreign financial assets (which are balance of payments debits, because the United States must pay foreigners for those assets) and increases in foreign claims on residents of the United States (which are balance of payments credits, because the United States receives payments when it sells assets overseas).

These data on increases in U.S. asset holdings abroad and foreign holdings of U.S. assets do not include holdings of *financial derivatives*, which are a class of assets that are more complicated than ordinary stocks and bonds, but have values that can depend on stock and bond values. (We will describe some specific derivative securities in the next chapter.) Starting in 2006, the U.S. Department of Commerce was able to assemble data on *net* cross-border derivative flows for the United States (U.S. net purchases of foreign-issued derivatives less foreign net purchases of U.S.-issued derivatives). Derivatives transactions enter the balance of payments accounts in the same way as do other international asset transactions.

According to Table 2-2, U.S.-owned assets abroad (other than derivatives) increased (on a net basis) by $225.40 billion in 2015. The figure is "on a net basis" because some U.S. residents bought foreign assets while others sold foreign assets they already owned, the difference between U.S. gross purchases and sales of foreign assets being $225.40 billion. In the same year (again on a net basis), the United States incurred new liabilities to foreigners equal to $395.23 billion. Some U.S. residents undoubtedly repaid foreign debts, but new borrowing from foreigners exceeded these repayments by $395.23 billion. The balance of U.S. purchases and sales of financial derivatives was −$25.39 billion: The United States acquired derivative claims on foreigners that were lower in value than the derivative claims on the U.S. that foreigners acquired. We calculate the balance on financial account (net financial flows) as $225.40 billion − $395.23 billion − $25.39 billion = −$195.23 billion. The negative value for net financial flows means that in 2015, the United States increased its net liability to foreigners (liabilities minus assets) by $195.23 billion.

Statistical Discrepancy

We come out with net financial flows of −$195.23 billion rather than the −$463.01 billion that we'd expected as a result of adding up the current and capital account balances. According to our data on trade and financial flows, the United States incurred $267.78 billion less in foreign debt than it actually needed to fund its current plus capital account deficit—because (−$195.23 billion) − (−$463.01 billion) = $267.78 billion. If every balance of payments credit automatically generates an equal counterpart debit and vice versa, how is this difference possible? The reason is that information about the offsetting debit and credit items associated with a given transaction may be collected from different sources. For example, the import debit that a shipment of DVD players from Japan generates may come from a U.S. customs inspector's report and the corresponding financial account credit from a report by the U.S. bank in which the check paying for the DVD players is deposited. Because data from different sources may differ in coverage, accuracy, and timing, the balance of payments accounts seldom balance in practice as they must in theory. Account keepers force the two sides to balance by adding to the accounts a *statistical discrepancy* item. For 2015, unrecorded (or misrecorded) international transactions generated a balancing accounting credit of $267.78 billion—the difference between the recorded net financial flows and the sum of the recorded current and capital accounts.

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We have no way of knowing exactly how to allocate this discrepancy among the current, capital, and financial accounts. (If we did, it wouldn't be a discrepancy!) The financial account is the most likely culprit, since it is notoriously difficult to keep track of the complicated financial trades between residents of different countries. But we cannot conclude that net financial flows were $267.78 billion lower than recorded because the current account is also highly suspect. Balance of payments accountants consider merchandise trade data relatively reliable, but data on services are not. Service transactions such as sales of financial advice and computer programming assistance may escape detection. Accurate measurement of international interest and dividend receipts can be particularly difficult.

Official Reserve Transactions

Although there are many types of financial account transactions, one type is important enough to merit separate discussion. This type of transaction is the purchase or sale of official reserve assets by central banks.

An economy's **central bank** is the institution responsible for managing the supply of money. In the United States, the central bank is the Federal Reserve System. **Official international reserves** are foreign assets held by central banks as a cushion against national economic misfortune. At one time, official reserves consisted largely of gold, but today, central banks' reserves include substantial foreign financial assets, particularly U.S. dollar assets such as Treasury bills. The Federal Reserve itself holds only a small level of official reserve assets other than gold; its own holdings of U.S. dollar assets are not considered international reserves.

Central banks often buy or sell international reserves in private asset markets to affect macroeconomic conditions in their economies. Official transactions of this type are called **official foreign exchange intervention.** One reason why foreign exchange intervention can alter macroeconomic conditions is that it is a way for the central bank to inject money into the economy or withdraw it from circulation. We will have much more to say later about the causes and consequences of foreign exchange intervention.

Government agencies other than central banks may hold foreign reserves and intervene officially in exchange markets. The U.S. Treasury, for example, operates an Exchange Stabilization Fund that at times has played an active role in market trading. Because the operations of such agencies usually have no noticeable impact on the money supply, however, we will simplify our discussion by speaking (when it is not too misleading) as if the central bank alone holds foreign reserves and intervenes.

When a central bank purchases or sells a foreign asset, the transaction appears in its country's financial account just as if the same transaction had been carried out by a private citizen. A transaction in which the central bank of Japan (the Bank of Japan) acquires dollar assets might occur as follows: A U.S. auto dealer imports a Nissan sedan from Japan and pays the auto company with a check for $20,000. Nissan does not want to invest the money in dollar assets, but it so happens that the Bank of Japan is willing to give Nissan Japanese money in exchange for the $20,000 check. The Bank of Japan's international reserves rise by $20,000 as a result of the deal. Because the Bank of Japan's dollar reserves are part of total Japanese assets held in the United States, the latter rise by $20,000. This transaction therefore results in a $20,000 credit in the U.S. financial account, the other side of the $20,000 debit in the U.S. current account due to the import of the car.[11]

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[11] To test your understanding, see if you can explain why the same sequence of actions causes a $20,000 improvement in Japan's current account and a $20,000 increase in its net financial flows.

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Table 2-2 shows that in 2015, U.S. official reserve assets fell by $6.29 billion. As shown in the table, foreign central banks sold $98.10 billion of the U.S. reserves they previously held. The net increase in U.S. official reserves *less* the increase in foreign official reserve claims on the United States is the level of net central bank financial flows, which stood at −$6.29 billion − (−$98.10) = $91.81 billion in 2015.

You can think of this $91.81 billion net central bank financial flow as measuring the degree to which monetary authorities in the United States and abroad generated additional net U.S. claims on foreigners—which help to finance the U.S. current account deficit when the number is negative, but increase the need for private foreign financing when it is positive. In the example above, the Bank of Japan, by acquiring a $20,000 U.S. bank deposit, indirectly finances an American import of a $20,000 Japanese car. The level of net central bank financial flows is called the **official settlements balance** or (in less formal usage) the **balance of payments.** This balance is the sum of the current account and capital account balances, less the nonreserve portion of the financial account balance, and it indicates the role of central banks' official reserve transactions in offsetting the current account balance. Thus, the U.S. balance of payments in 2015 was $91.81 billion.

The balance of payments played an important historical role as a measure of disequilibrium in international payments, and for many countries it still plays this role. A negative balance of payments (a deficit) may signal a crisis, for it means that a country is running down its international reserve assets or incurring debts to foreign monetary authorities. If a country faces the risk of being suddenly cut off from foreign loans, it will want to maintain a "war chest" of international reserves as a precaution. Many developing countries, in particular, are in this position (see Chapter 11).

Like any summary measure, however, the balance of payments must be interpreted with caution. To return to our running example, the Bank of Japan's decision to expand its U.S. bank deposit holdings by $20,000 swells the measured U.S. balance of payments deficit by the same amount. Suppose the Bank of Japan instead places its $20,000 with Barclays Bank in London, which in turn deposits the money with Citibank in New York. The United States incurs an extra $20,000 in liabilities to *private* foreigners in this case, and the U.S. balance of payments deficit does not rise. But this "improvement" in the balance of payments is of little economic importance: It makes no real difference to the United States whether it borrows the Bank of Japan's money directly or through a London bank.



CASE STUDY The Assets and Liabilities of the World's Biggest Debtor

We saw earlier that the current account balance measures the flow of new net claims on foreign wealth that a country acquires by exporting more goods and services than it imports. This flow is not, however, the only important factor that causes a country's net foreign wealth to change. In addition, changes in the market price of wealth previously acquired can alter a country's net foreign wealth. When Japan's stock market lost three-quarters of its value over the 1990s, for example, American and European owners of Japanese shares saw the value of their claims on Japan plummet, and Japan's net *foreign* wealth increased as a result. Exchange

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rate changes have a similar effect. When the dollar depreciates against foreign currencies, for example, foreigners who hold dollar assets see their wealth fall when measured in their home currencies.

The Bureau of Economic Analysis (BEA) of the U.S. Department of Commerce, which oversees the vast job of data collection behind the U.S. national income and balance of payments statistics, reports annual estimates of the net international investment position of the United States—the country's foreign assets less its foreign liabilities. Because asset price and exchange rate changes alter the dollar values of foreign assets and liabilities alike, the BEA must adjust the values of existing claims to reflect such capital gains and losses in order to estimate U.S. net foreign wealth. These estimates show that at the end of 2015, the United States had a *negative* net foreign wealth position far greater than that of any other country.

Until 1991, foreign direct investments such as foreign factories owned by U.S. corporations were valued at their historical, that is, original, purchase prices. Now the BEA uses two different methods to place current values on foreign direct investments: the *current cost* method, which values direct investments at the cost of buying them today, and the *market value* method, which is meant to measure the price at which the investments could be sold. These methods can lead to different valuations because the cost of replacing a particular direct investment and the price it would command if sold on the market may be hard to measure. (The net foreign wealth data graphed in Figure 2-2 are current cost estimates, which are believed to be more accurate.)

Table 2-3 reproduces the BEA's account of how it made its valuation adjustments to find the U.S. net IIP at the end of 2015. This "headline" estimate values direct investments at current cost. Starting with its estimate of 2014 net foreign wealth (−$7,046.1 billion), the BEA added the amount of the 2015 U.S. net financial flow of −$195.2 billion—recall the figure reported in Table 2-2. Then the BEA adjusted the values of previously held assets and liabilities for various changes in their dollar prices. As a result of these valuation changes, U.S. net foreign wealth fell by an amount greater than the $195.2 billion in new net borrowing from foreigners—in fact, U.S. net foreign wealth declined by $234.5 billion. The BEA's 2015 estimate of U.S. net foreign wealth, therefore, was −$7,280.6 billion.

This debt is larger than the total foreign debt owed by all the Central and Eastern European countries, which was about $822 billion in 2014. To put these figures in perspective, however, it is important to realize that the U.S. net foreign debt amounted to about 40 percent of its GDP, while the foreign liability of Hungary, Poland, Romania, and the other Central and Eastern European debtors was about 55 percent of their collective GDP! Thus, the U.S. external debt represents a lower domestic income drain.

Changes in exchange rates and securities prices have the potential to change the U.S. net foreign debt sharply, however, because the *gross* foreign assets and liabilities of the United States have become so large in recent years. Figure 2-3 illustrates this dramatic trend. In 1976, U.S. foreign assets stood at only 20 percent of U.S. GDP and liabilities at 15 percent (making the United States a net foreign creditor in the amount of roughly 5 percent of its GDP). In 2015, however, the country's foreign assets amounted to roughly 130 percent of GDP and its liabilities to roughly 171 percent. The tremendous growth in these stocks of wealth reflects the rapid globalization of financial markets in the late 20th century, a phenomenon we will discuss further in Chapter 9.

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TABLE 2-3 Change in the Yearend U.S. Net International Investment Position (billions of dollars)

Line	Type of investment	Yearend position, 2014 [r]	Change in position in 2015						Yearend position, 2015 [r]
			Total	Attributable to:					
				Financial-account transactions	Other changes in position				
					Total	Price changes	Exchange-rate changes [1]	Changes in volume and valuation n.i.e. [2]	
1	U.S. net international investment position (line 4 less line 35)	−7,046.1	−234.5	−195.2	−39.3	(4)	(4)	(4)	−7,280.6
2	Net international investment position excluding financial derivatives (line 5 less line 36)	−7,131.7	−206.2	−169.8	−36.4	781.4	−1,051.5	233.7	−7,337.9
3	Financial derivatives other than reserves, net (line 6 less line 37) 3	85.5	−28.3	−25.4	−2.9	(4)	(4)	(4)	57.2
4	U.S. assets	24,717.5	−1,376.8	(3)	(3)	(3)	(3)	(3)	23,340.8
5	Assets excluding financial derivatives (sum of lines 7, 10, 21, and 26)	21,503.4	−558.0	225.4	−783.4	220.4	−1,141.5	137.7	20,945.4
6	Financial derivatives other than reserves, gross positive fair value (line 15)	3,214.1	−818.8	(3)	(3)	(3)	(3)	(3)	2,395.4
	By functional category:								
7	Direct investment at market value	7,133.1	−154.8	348.6	−503.4	−64.7	−449.3	10.5	6,978.3
8	Equity	6,045.1	−234.0	316.3	−550.3	−64.7	−449.3	−36.3	5,811.1
9	Debt instruments	1,088.1	79.2	32.3	46.9	....	....	46.9	1,167.2
10	Portfolio investment	9,704.3	−98.1	154.0	−252.1	323.2	−631.3	56.0	9,606.2
11	Equity and investment fund shares	6,770.6	57.6	202.6	−145.0	357.9	−562.9	60.1	6,828.2
12	Debt securities	2,933.6	−155.7	−48.6	−107.1	−34.6	−68.4	−4.1	2,777.9
13	Short term	447.2	39.1	42.5	−3.4	....	−3.4	0.0	486.2
14	Long term	2,486.4	−194.7	−91.1	−103.6	−34.6	−65.0	−4.1	2,291.7
15	Financial derivatives other than reserves, gross positive fair value	3,214.1	−818.8	(3)	(3)	(3)	(3)	(3)	2,395.4
16	Over-the-counter contracts	3,144.0	−797.3	(3)	(3)	(3)	(3)	(3)	2,346.7
17	Single-currency interest rate contracts	2,451.1	−643.7	(3)	(3)	(3)	(3)	(3)	1,807.4
18	Foreign exchange contracts	415.4	−73.2	(3)	(3)	(3)	(3)	(3)	342.3
19	Other contracts	277.5	−80.4	(3)	(3)	(3)	(3)	(3)	197.1
20	Exchange-traded contracts	70.1	−21.5	(3)	(3)	(3)	(3)	(3)	48.6
21	Other investment	4,231.8	−254.5	−270.9	16.4	....	−54.7	71.1	3,977.3
22	Currency and deposits	1,785.5	−156.9	−194.4	37.5	....	−30.6	68.0	1,628.6
23	Loans	2,399.2	−95.3	−74.8	−20.5	....	−23.6	3.1	2,304.0
24	Insurance technical reserves	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
25	Trade credit and advances	47.0	−2.3	−1.7	−0.6	....	−0.6	0.0	44.7
26	Reserve assets	434.3	−50.7	−6.3	−44.4	−38.2	−6.2	0.0	383.6
27	Monetary gold	315.4	−38.2	0.0	−38.2	−38.2	....	0.0	277.2
28	Special drawing rights	51.9	−2.3	(*)	−2.3	....	−2.3	0.0	49.7
29	Reserve position in the International Monetary Fund	25.2	−7.6	−6.5	−1.1	....	−1.1	0.0	17.6
30	Other reserve assets	41.8	−2.7	0.2	−2.8	0.0	−2.8	0.0	39.1
31	Currency and deposits	19.0	−1.4	(*)	−1.4	....	−1.5	0.1	17.6
32	Securities	22.8	−1.2	0.2	−1.4	0.0	−1.4	−0.1	21.6
33	Financial derivatives	....	....	....	....	....	....	....	....
34	Other claims	0.0	0.0	0.0	0.0	....	0.0	0.0	0.0
35	U.S. liabilities	31,763.7	−1,142.3	(3)	(3)	(3)	(3)	(3)	30,621.4
36	Liabilities excluding financial derivatives (sum of lines 38, 41, and 56)	28,635.1	−351.8	395.2	−747.0	−561.0	−90.0	96.0	28,283.3
37	Financial derivatives other than reserves, gross negative fair value (line 50)	3,128.6	−790.5	(3)	(3)	(3)	(3)	(3)	2,338.1
	By functional category:								
38	Direct investment at market value	6,350.1	193.8	379.4	−185.7	−160.3	....	−25.3	6,543.8
39	Equity	4,884.1	95.2	301.1	−205.9	−160.3	....	−45.6	4,979.3
40	Debt instruments	1,466.0	98.6	78.3	20.2	....	....	20.2	1,564.5
41	Portfolio investment	16,919.8	−242.8	250.9	−493.7	−400.7	−57.0	−36.0	16,677.0
42	Equity and investment fund shares	6,642.5	−423.6	−178.3	−245.4	−187.8	....	−57.5	6,218.9
43	Debt securities	10,277.3	180.8	429.2	−248.4	−212.9	−57.0	21.5	10,458.1
44	Short term	911.8	43.4	45.8	−2.4	....	−2.4	0.0	955.2
45	Treasury bills and certificates	671.6	53.1	53.1	0.0	....	....	0.0	724.7
46	Other short-term securities	240.2	−9.7	−7.3	−2.4	....	−2.4	0.0	230.5
47	Long term	9,365.5	137.5	383.4	−246.0	−212.9	−54.6	21.5	9,503.0
48	Treasury bonds and notes	5,484.4	−61.0	−4.8	−56.3	−56.3	....	0.0	5,423.4
49	Other long-term securities	3,881.1	198.5	388.2	−189.7	−156.6	−54.6	21.5	4,079.6
50	Financial derivatives other than reserves, gross negative fair value	3,128.6	−790.5	(3)	(3)	(3)	(3)	(3)	2,338.1
51	Over-the-counter contracts	3,062.6	−771.5	(3)	(3)	(3)	(3)	(3)	2,291.1
52	Single-currency interest rate contracts	2,398.8	−643.4	(3)	(3)	(3)	(3)	(3)	1,755.4
53	Foreign exchange contracts	393.6	−49.6	(3)	(3)	(3)	(3)	(3)	344.0
54	Other contracts	270.2	−78.5	(3)	(3)	(3)	(3)	(3)	191.7
55	Exchange-traded contracts	66.0	−19.0	(3)	(3)	(3)	(3)	(3)	47.0
56	Other investment	5,365.2	−302.7	−235.1	−67.6	....	−33.0	−34.6	5,062.5

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Line	Type of investment	Yearend position, 2014 r	Change in position in 2015						Yearend position, 2015 r
			Total	Financial-account transactions	Attributable to: Other changes in position				
					Total	Price changes	Exchange-rate changes [1]	Changes in volume and valuation n.i.e. [2]	
57	Currency and deposits	2,886.7	27.5	33.4	−5.9	....	−10.8	4.9	2,914.3
58	Loans	2,265.6	−342.0	−282.7	−59.3	....	−19.7	−39.5	1,923.6
59	Insurance technical reserves	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.
60	Trade credit and advances	161.7	13.9	14.2	−0.3	....	−0.3	0.0	175.6
61	Special drawing rights allocations	51.2	−2.2	0.0	−2.2	....	−2.2	0.0	48.9

r Revised n.a. Not available .... Not applicable (*) Value between zero and +/−$50 million

1 Represents gains or losses on foreign-currency-denominated assets and liabilities due to their revaluation at current exchange rates.

2 Includes changes due to year-to-year shifts in the composition of reporting panels and to the incorporation of more comprehensive survey results. Also includes capital gains and losses of direct investment affiliates and changes in positions that cannot be allocated to financial transactions, price changes, or exchange-rate changes.

3 Financial transactions and other changes in financial derivatives positions are available only on a net basis, which is shown on line 3; they are not separately available for gross positive fair values and gross negative fair values of financial derivatives.

4 Data are not separately available for price changes, exchange-rate changes, and changes in volume and valuation not included elsewhere.

Note: Details may not add to totals because of rounding.

Source: U.S. Bureau of Economic Analysis.

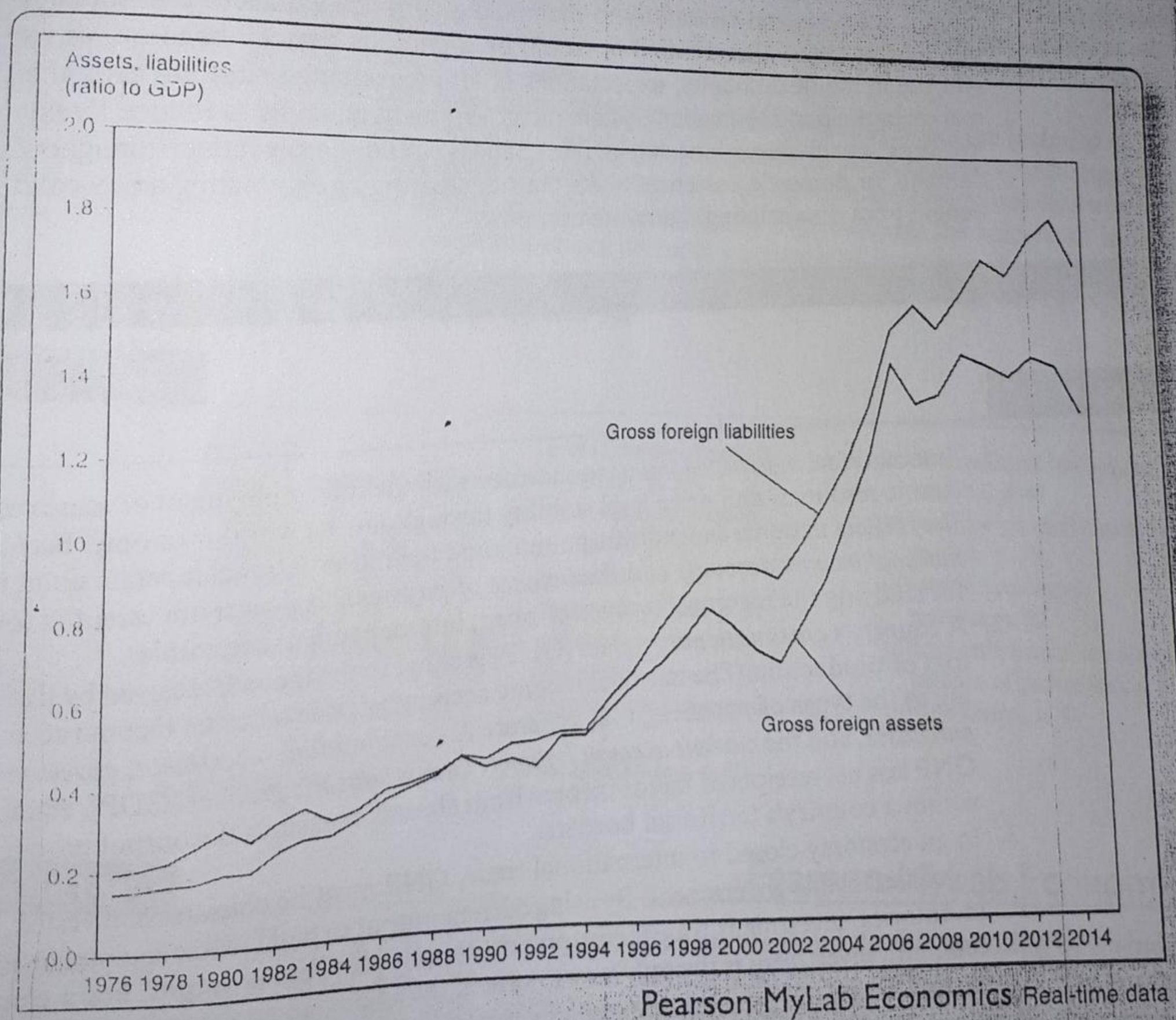


FIGURE 2-3

Gross Foreign Assets and Liabilities, 1976–2015

Since 1976, both the foreign assets and the liabilities of the United States have increased sharply. But liabilities have risen more quickly, leaving the United States with a substantial net foreign debt.

Source: U.S. Department of Commerce, Bureau of Economic Analysis, June 2016.

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Think about how gross wealth positions of this magnitude amplify the effects of exchange rate changes, however. Suppose 70 percent of U.S. foreign assets are denominated in foreign currencies, but all U.S. liabilities to foreigners are denominated in dollars (these are approximately the correct numbers). Because in 2015 U.S. GDP was around $18 trillion, a 10 percent depreciation of the dollar would leave U.S. liabilities unchanged but would increase U.S. assets (measured in dollars) by $0.1 \times 0.7 \times 1.30 = 9.1$ percent of GDP, or about $1.6 trillion. This number is approximately 3.4 times the U.S. current account deficit of 2015! Indeed, due to sharp movements in exchange rates and stock prices, the U.S. economy lost about $800 billion in this way between 2007 and 2008 and gained a comparable amount between 2008 and 2009 (see Figure 2-2). The corresponding redistribution of wealth between foreigners and the United States would have been much smaller back in 1976.

Does this possibility mean that policy makers should ignore their countries' current accounts and instead try to manipulate currency values to prevent large buildups of net foreign debt? That would be a perilous strategy because, as we will see in the next chapter, expectations of future exchange rates are central to market participants' behavior. Systematic government attempts to reduce foreign investors' wealth through exchange rate changes would sharply reduce foreigners' demand for domestic currency assets, thus decreasing or eliminating any wealth benefit from depreciating the home currency.

SUMMARY

1. International *macroeconomics* is concerned with the full employment of scarce economic resources and price level stability throughout the world economy. Because they reflect national expenditure patterns and their international repercussions, the *national income accounts* and the *balance of payments accounts* are essential tools for studying the macroeconomics of open, interdependent economies.
2. A country's *gross national product* (GNP) is equal to the income received by its factors of production. The national income accounts divide national income according to the types of spending that generate it: *consumption*, *investment*, *government purchases*, and the *current account balance*. *Gross domestic product* (GDP), equal to GNP less net receipts of factor income from abroad, measures the output produced within a country's territorial borders.
3. In an economy closed to international trade, GNP must be consumed, invested, or purchased by the government. By using current output to build plant, equipment, and inventories, investment transforms present output into future output. For a closed economy, investment is the only way to save in the aggregate, so the sum of the saving carried out by the private and public sectors, *national saving*, must equal investment.
4. In an open economy, GNP equals the sum of consumption, investment, government purchases, and net exports of goods and services. Trade does not have to be balanced if the economy can borrow from and lend to the rest of the world. The difference between the economy's exports and imports, the current account balance, equals the difference between the economy's output and its total use of goods and services.
5. The current account also equals the country's net lending to foreigners. Unlike a closed economy, an open economy can save through domestic *and* foreign

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investments. National saving therefore equals domestic investment plus the current account balance. The current account is closely related to the change in the *net international investment position*, though usually not equal to that change because of fluctuations in asset values not recorded in the national income and product accounts.

6. Balance of payments accounts provide a detailed picture of the composition and financing of the current account. All transactions between a country and the rest of the world are recorded in the country's balance of payments accounts. The accounts are based on the convention that any transaction resulting in a payment to foreigners is entered as a debit while any transaction resulting in a receipt from foreigners is entered as a credit.
7. Transactions involving goods and services appear in the current account of the balance of payments, while international sales or purchases of *assets* appear in the *financial account*. The *capital account* records mainly nonmarket asset transfers and tends to be small for the United States. The sum of the current and capital account balances must equal the financial account balance (net financial flows). This feature of the accounts reflects the fact that discrepancies between export earnings and import expenditures must be matched by a promise to repay the difference, usually with interest, in the future.
8. International asset transactions carried out by *central banks* are included in the financial account. Any central bank transaction in private markets for foreign currency assets is called *official foreign exchange intervention.* One reason intervention is important is that central banks use it as a way to change the amount of money in circulation. A country has a deficit in its *balance of payments* when it is running down its *official international reserves* or borrowing from foreign central banks; it has a surplus in the opposite case.

KEY TERMS

asset, p. 44
balance of payments accounting, p. 33
capital account, p. 44
central bank, p. 50
consumption, p. 36
current account balance, p. 38
financial account, p. 44
government budget deficit, p. 42
government purchases, p. 37
gross domestic product (GDP), p. 35
gross national product (GNP), p. 33
investment, p. 36
macroeconomics, p. 31
microeconomics, p. 31
national income, p. 34
national income accounting, p. 33
national saving, p. 40
net international investment position, p. 40
official foreign exchange intervention, p. 50
official international reserves, p. 50
official settlements balance (or balance of payments), p. 51
private saving, p. 42

PROBLEMS

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1. We stated in this chapter that GNP accounts avoid double counting by including only the value of *final* goods and services sold on the market. Should the measure of imports used in the GNP accounts therefore be defined to include only imports of final goods and services from abroad? What about exports?
2. Equation (2-2) tells us that to reduce a current account deficit, a country must increase its private saving, reduce domestic investment, or cut its government budget deficit. Nowadays, some people recommend restrictions on imports from China (and other countries) to reduce the American current account deficit. How would higher U.S. barriers to imports affect its private saving, domestic investment, and

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government deficit? Do you agree that import restrictions would necessarily reduce a U.S. current account deficit?

3. Explain how each of the following transactions generates two entries—a credit and a debit—in the Japanese balance of payments accounts, and describe how each entry would be classified:
   a. A Japanese investor buys a share of a Thailand stock company, paying by writing a check on an account with a Singapore bank.
   b. A Japanese investor buys a share of a Thailand stock company, paying the seller with a check on a Japanese bank.
   c. The Malaysian government carries out an official foreign exchange intervention in which it uses Yen held in a Japanese bank to buy ringgit (Malaysian currency) from its citizens
   d. A tourist from Kyoto buys a meal at an expensive restaurant in Bangkok, paying with a traveler's check.
   e. A saké brewery in Kobe contributes a case of saké bottles for a Paris wine-tasting event.
   f. A Japanese-owned factory in Germany uses local earnings to buy additional machinery.
4. A Mexican travels to Brazil to buy a gemstone which costs 3000 Real (Brazilian currency). The Brazilian company that sells the gemstone then deposits the 3000 Real in its account in a Panama bank. How would these transactions show up in the balance of payments accounts of Mexico and Brazil? What if the Mexican pays cash for the gemstone?
5. The nation of Pecunia had a current account deficit of $1 billion and a nonreserve financial account surplus of $500 million in 2017.
   a. What was the balance of payments of Pecunia in that year? What happened to the country's net foreign assets?
   b. Assume that foreign central banks neither buy nor sell Pecunian assets. How did the Pecunian central bank's foreign reserves change in 2017? How would this official intervention show up in the balance of payments accounts of Pecunia?
   c. How would your answer to (b) change if you learned that foreign central banks had purchased $600 million of Pecunian assets in 2017? How would these official purchases enter foreign balance of payments accounts?
   d. Draw up the Pecunian balance of payments accounts for 2017 under the assumption that the event described in (c) occurred in that year.
6. Can you think of reasons why a government might be concerned about a large current account deficit or surplus? Why might a government be concerned about its official settlements balance (that is, its balance of payments)?
7. Do the data on the European Central Bank, which set up the official settlements balance for the Eurozone, give an accurate picture of the extent to which foreign central banks buy and sell euros in currency markets?
8. Is it possible for a country to have a current account deficit at the same time it has a surplus in its balance of payments? Explain your answer, using hypothetical figures for the current and nonreserve financial accounts. Be sure to discuss the possible implications for official international reserve flows.
9. Suppose the South Africa net foreign debt is 50 percent of its GDP and foreign assets and liabilities pay an interest rate of 6 percent per year. What would be the drain on SA GDP (as a percentage) from paying interest on the net foreign debt? Do you think this is a large number? What if the net foreign debt were 100 percent of GDP? At what point do you think a country's government should become worried about the size of its foreign debt?

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10. In 2015, in the balance of payment of Switzerland, the labor income in the primary income balance shows a deficit of 21 billion CHF (Swiss Franc), which represents almost 30 percent of the net account surplus (71 Billion CHF). How can you explain the size of this deficit?
11. Return to the example in this chapter's final Case Study of how a 10 percent dollar depreciation affects U.S. net foreign wealth (pp. 51–55). Show the size of the effect on *foreigners'* net foreign claims on the United States measured in dollars (as a percent of U.S. GDP).
12. We mentioned in the chapter that capital gains and losses on a country's net foreign assets are not included in the national income measure of the current account. How would economic statisticians have to modify the national income identity [equation (2-1)] if they wish to include such gains and losses as part of the definition of the current account? In your opinion, would this make sense? Why do you think this is not done in practice?
13. The international investment position of the European Union (28 countries) is negative of €2,557.4 billion in 2015, from €3,337.8 billion in 2014. The breakdown by country is as follows:

	Assets	Liabilities	Net	Net (% GDP)
Belgium	1,972.5	1,719.2	253.3	61.9
Bulgaria	37.9	64.7	−26.8	−60.7
Czech Republic	169.6	221.3	−51.7	−31.0
Denmark	793.8	682.1	111.7	42.0
Germany	7,900.6	6,418.6	1,482.0	49.0
Estonia	28.4	36.7	−8.3	−40.5
Ireland	4,747.4	5,279.5	−532.1	−208.0
Greece	243.4	477.5	−234.1	−133.0
Spain	1,640.9	2,615.8	−974.9	−90.2
France	6,422.1	6,780.2	−358.1	−16.4
Croatia	30.7	64.8	−34.1	−77.6
Italy	2,362.8	2,758.4	−395.6	−24.2
Cyprus	207.5	230.1	−22.5	−129.2
Latvia	30.5	44.9	−14.5	−59.3
Lithuania	22.9	39.7	−16.8	−45.2
Luxembourg	9,470.5	9,454.0	16.5	31.6
Hungary	269.7	339.5	−69.8	−64.1
Malta	223.1	219.1	4.0	45.4
Netherlands	7,583.2	7,171.8	411.4	60.8
Austria	892.9	877.4	15.5	4.6
Poland	209.7	471.2	−261.6	−61.1
Portugal	333.0	529.2	−196.2	−109.4
Romania	57.7	138.2	−80.6	−50.2
Slovenia	40.3	55.1	−14.8	−38.5
Slovakia	51.8	106.1	−54.3	−69.6
Finland	697.8	705.8	−8.0	−3.8
Sweden	1,299.1	1,305.7	−6.6	−1.5
United Kingdom	13,443.1	13,809.8	−366.6	−14.2

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What are the main creditor countries? What are the main debtor countries? Explain. (Hint: refer to the analysis on Eurostat analysis: http://ec.europa.eu/eurostat/statistics-explained/index.php/Economy_and_finance)

FURTHER READINGS

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6. Disequilibrium in Balance of Payments

A disequilibrium in the *BOP* of a country may be either a deficit or a surplus. A deficit or surplus in *BOP* of a country appears when its autonomous receipts (credits) do not match its autonomous payments (debits). If autonomous credit receipts exceed autonomous debit payments, there is a surplus in the *BOP* and the disequilibrium is said to be *favourable*. On the other hand, if autonomous debit payments exceed autonomous credit receipts, there is a *deficit* in the *BOP* and the disequilibrium is said to be unfavourable or adverse.[5]

Causes of Disequilibrium

There are many factors that may lead to a *BOP* deficit or surplus:

1. **Temporary Changes (or Disequilibrium).** There may be a temporary disequilibrium caused by random variations in trade, seasonal fluctuations, the effects of weather on agricultural production, etc. Deficits or surpluses arising from such temporary causes are expected to correct themselves within a short time.

2. **Fundamental Disequilibrium.** Fundamental disequilibrium refers to a persistent and long-run *BOP* disequilibrium of a country. It is a chronic *BOP* deficit, according to *IMF*. It is caused by such dynamic factors as : (1) Changes in consumer tastes within the country or abroad which reduce the country's exports and increase its imports. (2) Continuous fall in the country's foreign exchange reserves due to supply inelasticities of exports and excessive demand for foreign goods and services. (3) Excessive capital outflows due to massive

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imports of capital goods, raw materials, essential consumer goods, technology and external indebedness. (4) Low competitive strength in world markets which adversly affects exports. (5) Inflationary pressures within the economy which make exports dearer.

3. Structural Changes (or Disequilibrium). Structural changes bring about disequilibrium in *BOP* over the long run. They may result from the following factors: (*a*) Technological changes in methods of production of products in domestic industries or in the industries of other countries. They lead to changes in costs, prices and quality of products. (*b*) Import restrictions of all kinds bring about disequilibrium in *BOP*. (*c*) Deficit in *BOP* also arises when a country suffers from deficiency of resources which it is required to import from other countries. (*d*) Disequilibrium in *BOP* may also be caused by changes in the supply or direction of long-term capital flows. More and regular flow of long-term capital may lead to *BOP* surplus, while an irregular and short supply of capital brings *BOP* deficit.

4. Changes in Exchange Rates. Changes in foreign exchange rate in the form of overvaluation or undervaluation of foreign currency lead to *BOP* disequilibrium. When the value of currency is higher in relation to other currencies, it is said to be overvalued. Opposite is the case of an undervalued currency. Overvaluation of the domestic currency makes foreign goods cheaper and exports dearer in foreign countries. As a result, the country imports more and exports less of goods. There is also outflow of capital. This leads to unfavourable *BOP*. On the contrary, undervaluation of the currency makes *BOP* favourable for the country by encouraging exports and inflow of capital and reducing imports.

5. The distinction between deficit and disequilibrium in *BOP* should be clearly understood. The former refers tounfavourable or adverse *BOP* as explained above whereas disequilibrium refers to a deficit or surplus in *BOP* .

5. Cyclical Fluctuations (or Disequilibrium). Cyclical fluctuations in business activity also lead to *BOP* disequilibrium. When there is depression in a country, volumes of both exports and imports fall

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drastically in relation to other countries. But the fall in exports may be more than that of imports due to decline in domestic production. Therefore, there is an adverse *BOP* situation. On the other hand, when there is boom in a country in relation to other countries, both exports and imports may increase. But there can be either a surplus or deficit in *BOP* situation depending upon whether the country exports more than imports or imports more than exports. In both the cases, there will be disequilibrium in BOP.

6. **Changes in National Income.** Another cause is the change in the country's national income. If the national income of a country increases, it will lead to an increase in imports thereby creating a deficit in its balance of payments, other things remaining the same. If the country is already at full employment level, an increase in income will lead to inflationary rise in prices which may increase its imports and thus bring disequilibrium in the balance of payments.

7. **Price Changes.** Inflation or deflation is another cause of disequilibrium in the balance of payments. If there is inflation in the country, prices of exports increase. As a result, exports fall. At the same time, the demand for imports increase. Thus increase in export prices leading to decline in exports and rise in imports results in adverse balance of payments.

8. **Stage of Economic Development.** A country's balance of payments also depends on its stage of economic development. If a country is developing, it will have a deficit in its balance of payments because it imports raw materials, machinery, capital equipment, and services associated with the development process and exports primary products. The country has to pay more for costly imports and gets less for its cheap exports. This leads to disequilibrium in its balance of payments.

9. **Capital Movements.** Borrowings and lendings or movements of capital by countries also result in disequilibrium in BOP. A country which gives loans and grants on a large scale to other countries has a deficit in its *BOP* on capital account. If it is also importing more, as is the case with the USA, it will have chronic deficit. On the other

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hand, a developing country brorrowing large funds from other countries and international institutions may have a favourable BOP. But such a possibility is remote because these countries usually import huge quanties of food, raw materials, capital goods, etc. and export primary products. Such borrowings simply help in reducing *BOP* deficit.

10. Political Conditions. Political condition of a country is another cause of disequilibrium in BOP. Political instability in a country creates uncertainty among foreign investors which leads to the outflow of capital and retards its inflow. This causes disequilibrium in *BOP* of the country. Disequilibrium in *BOP* also occurs in the event of war or fear of war with some other country.

Implications of Disequilibrium

A disequilibrium in the balance of payments whether a deficit or surplus has important implications for a country.

A deficit in the combined current and capital accounts is regarded as undesirable for the country. This is because such a deficit has to be covered by borrowing from abroad or attracting foreign exchange or capital from abroad. This may require paying high interest rates. There is also the danger of withdrawing money by foreigners, as happened in the case of the Asian crisis in the late 1990s. An alternative may be to draw on the reserves of the country which may also lead to a financial crises. Moreover, the reserves of a country being limited, they can be used to pay for *BOP* deficit upto a limit.

But the above analysis of a combined current and capital account deficit is not correct in practice. The reason being that a current account deficit is the same thing as a capital account surplus. However, it is beneficial for a country to have a current account deficit even if it equals capital account surplus in BOP. In the short-run, the country may benefit from a higher level of consumption through import of goods and consequently a higher standard of living. But the excess of imports over exports may be financed by

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foreign investments in the country. These may lead to increased production, employment and income in the country. In the long-run, foreign investors may purchase large assets in the country and thus adversely affect domestic industry as is the case with MNCs (multinational corporations).

The current account deficit in *BOP* of a country may have either good or bad effects depending on the nature of an economy.

Take a country where domestic industries are rapidly growing and it has current account *BOP* deficit. These industries offer a high rate of return on their investment. This would, in return, attract foreign investments. As a result, the country would have a capital account surplus due to the inflow of capital and a current account deficit. This current account deficit is good for the economy. No doubt, the external debt of the country increases, but this debt is being utilised to finance the rapid growth of the economy. The real burden of this debt will be very low because it can be repaid out of higher income in the future.

On the contrary, a country having an inefficient and unproductive domestic industry will be adversely affected by its current account *BOP* deficit. The country borrows from abroad to finance the excess of spending over consumption. To attract foreign borrowings, the country will have to pay high interest rates. These will increase the money burden of the debt. The real burden of the debt will also increase because of the low productive capacity of domestic industries. If the current consumption is being financed by foreign borrowings, the wealth of the economy will decline. This, in turn, will lead to either a reduction in domestic expenditure or a change in government policy so as to control the rising debt.

On the other hand, if foreign borrowings are being used to finance real investment, the current account *BOP* deficit will be beneficial for the economy. A higher rate of return on real investment than the interest on foreign borrowings would increase the country's wealth over time through rise in its national income. Thus a current account *BOP* deficit is not always undesirable for a country.

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7. Measures to Correct Deficit in Balance of Payments[6]

When there is a deficit in the balance of payments of a country, adjustment is brought about automatically through price and income changes or by adopting certain policy measures like export promotion, monetary and fiscal policies, devaluation and direct controls. We study these as follows:

1. Adjustment through Exchange Depreciation (Price Effect)

Under flexible exchange rates, the disequilibrium in the balance of payments is automatically solved by the forces of demand and supply for foreign exchange. An exchange rate is the price of a currency which is determined, like any other commodity, by demand and supply. "The exchange rate varies with varying supply and demand conditions, but it is always possible to find an equilibrium exchange rate which clears the foreign exchange market and creates external equilibrium."[7] This is automatically achieved by depreciation of a country's currency in case of deficit in its balance of payments. Depreciation of a currency means that its relative value decreases. Depreciation has the effect of encouraging exports and discouraging imports. When exchange depreciation takes place, foreign prices are translated into domestic prices. Suppose the dollar depreciates in relation to the pound. It means, that the price of dollar falls in relation to the pound in the foreign exchange market. This leads to the lowering of the prices of U.S. exports in Britain and raising of the prices of British imports in the U.S. When import prices are higher in the U.S., the Americans will purchase less goods from the Britishers. On the other hand, lower prices of U.S. exports will increase exports and diminish imports, thereby bringing equilibrium in the balance of payments.

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6. if the question relates to disequilibrium in *BOP* , then measures opposite to whatever are given in the paras shouldalso be explained relating to *BOP* surplus, such as appreciation, revaluation, etc. The best answer would be to writeat the end of each para : The opposite measures would be adopted in case of *BOP* surplus. Mention the measures inthe dotted portion.

7. Bo Sodersten, *International Economics*, p. 215:

2. DEVALUATION OR EXPENDITURE-SWITCHING POLICY

Devaluation raises the domestic price of imports and reduces the foreign price of exports of a country devaluing its currency in relation to the currency of another country. Devaluation is referred to as expenditure switching policy because it switches expenditure from imported to domestic goods and services. When a country devalues its currency, the price of foreign currency increases which makes imports dearer and exports cheaper. This causes expenditures to be switched from foreign to domestic goods as the country's exports rise and the country produces more to meet the domestic and foreign demand for goods with reduction in imports. Consequently, the balance of payments deficit is eliminated.

3. DIRECT CONTROLS

To correct disequilibrium in the balance of payments, government also adopts direct controls which aim at limiting the volume of imports.The government restricts the import of undesirable or unimportant items by levying heavy import duties, fixation of quotas, etc. At the same time, it may allow imports of essential goods duty free or at lower import duties, or fix liberal import quotas for them. For instance, the government may allow free entry of capital goods, but impose heavy import duties on luxuries. Import quotas are also fixed and the importers are required to take licenses from the authorities in order to import certain essential commodities in fixed quantities. In these ways, imports are reduced in order to correct an adverse balance of payments. The government also imposes

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exchange controls. Exchange controls have a dual purpose. They restrict imports and also control and regulate the foreign exchange. With reduction in imports and control of foreign exchange, visible and invisible imports are reduced. Consequently, an adverse balance of payment is corrected.

4. Adjustment through Capital Movements

A country can use capital imports to correct a deficit in its balance of payments. A deficit can be financed by capital inflows. When capital is perfectly mobile within countries, a small rise in the domestic rate of interest brings a large inflow of capital. The balance of payments is said to be in equilibrium when the domestic interest rate equals the world rate. If the domestic interest rate is higher than the world rate, there will be capital inflows and the balance of payments deficit is corrected.

5. Adjustment through Income Changes

Given the foreign exchange rate and prices in a country, an increase in the value of exports, causes an increase in the incomes of all persons associated with the export industries. These, in turn, create demand for other goods and services within the country. This will raise the incomes of persons engaged in the latter industries and services. This process will continue and the national income increases by the value of the multiplier.

6. Stimulation of Exports and Import Substitutes

A deficit in the balance of payments can also be corrected by encouraging exports. Exports can be encouraged by producing quality products, by increasing exports through increased production and productivity, and by better marketing. They can also be increased by a policy of import substitution. It means that the country produces those goods which it imports. In the beginning, imports are reduced but in the long run exports of such goods start. An increase in exports cause the national income to rise by many times through

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the operation of the foreign trade multiplier. The foreign trade multiplier expresses the change in income caused by a change in exports. Ultimately, the deficit in the balance of payments is removed when exports rise faster than imports.

7. EXPENDITURE-REDUCING POLICIES

A deficit in the balance of payments implies an excess of expenditure over income. To correct it, expenditure and income should be brought into equality. For this expenditure reducing monetary and fiscal policies are used. A contractionary or tight monetary policy relates to increase in interest rates to reduce money supply and a contractionary fiscal policy relates to reduction in government expenditure and or increase in taxes. Thus expenditure reducing policies reduce aggregate demand through higher taxes and interest rates, thereby reducing expenditure and output. The reduction in expenditure and output, in turn, reduces the domestic price level. This gives rise to switching of expenditure from foreign to domestic goods. Consequently, the country's imports are reduced and the balance of payments deficit is corrected.

EXERCISES

1. Enumerate the principle items in the balance of payments of a country. How can a deficit in the balance of payments be corrected?

2. "Balance of Payments always balances." Elucidate. But how do you explain disequilibrium in balance of payments.

3. Distinguish between balance of payments and balance of trade. How can an unfavourable balance of payments be corrected?

4. What are the causes of an adverse balance of payments? Give suggestions to remove an unfavourable balance of payments?

5. Distinguish between : *(a)* Balance of Current Account and Balance of Capital Account; *(b)* Autonomous Balance and Overall

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